Few-Shot Learning With Label Smoothing and Metric Space Optimization for Hyperspectral Image Classification

Dongmei Song, Fuhou Qin, Bin Wang, and Song Dai

***Abstract*— Due to the high operational complexity of manual sample labeling for hyperspectral images (HSIs), few-shot learning (FSL) has been introduced to cope with the lack of training samples in the field of HSI classification and achieved good results by virtue of its excellent performance. However, factors such as category labeling noise, category distinguishability in the metric space, and completeness of effective feature mining are still the primary considerations that significantly affect the stability and robustness of FSL. Therefore, this study proposes a framework of FSL with label smoothing and metric space optimization (LMFSL) for HSI classification. The framework first incorporates a category label smoothing strategy into FSL, which mitigates the effect of noise by constructing a novel category label smoothing module (CLSM), to reduce the confidence of the classifier. Meanwhile, the study also designs a metric space optimization module (MSOM), which prompts similar samples in the metric space to largely aggregate together by maximizing the intraclass similarity and minimizing the interclass similarity in a more flexible way, so as to improve the decision boundary of the model and enhance the recognition performance of the model. Furthermore, to achieve effective feature extraction in the context of a few labeled samples, a lightweight feature extractor, LFE-MAs, incorporating multiple attention mechanisms is designed to extract the HSI features with high efficiency and low computational cost. Experimental results on four public HSI datasets show that LMFSL outperforms other state-of-the-art methods in terms of classification accuracy with limited labeled samples and has lower computational complexity.**

***Index Terms*— Few-shot learning (FSL), hyperspectral image (HSI) classification, label smoothing, metric space optimization, multiple attention mechanisms.**

I. Introduction

As an important technique in remote sensing, hyperspectral imaging is used to image target features by utilizing hundreds of continuous bands. In a hyperspectral image (HSI), each pixel can be regarded as a high-dimensional vector, which contains rich spectral information. Up to now, HSI has been widely utilized in various fields [1], [2], [3], [4], [5]. HSI classification is a pixel-level classification method based on HSI features, and its correct classification is a prerequisite for the application in various fields [6].

Early HSI classification research primarily focused on shallow feature extraction techniques, such as principal component analysis (PCA) [7] and linear discriminant analysis (LDA) [8], as well as classifiers including random forests (RFs) [9], extreme learning machines (ELMs) [10], K-nearest neighbors (KNNs) [11], and support vector machines (SVMs) [12]. These methods are able to achieve prediction of new samples by extracting certain recognizable features from the training samples, while exhibiting relatively robust performance in the process. However, these methods also have obvious limitations: one is that most of them focus only on the features of a single pixel, ignoring the potential correlations between the pixel and its surrounding neighborhoods, thus making it difficult to capture more complex semantic information. The second is that since feature extraction and feature classification are two independent processes, there is a lack of effective integrating mechanisms to achieve the consistent optimization between the two in these early traditional methods.

Deep learning, as a powerful end-to-end feature extraction method, has been widely applied in HSI classification with good results [13]. Examples include stacked autoencoders (SAEs) [14], deep belief networks (DBNs) [15], and recurrent neural networks (RNNs) [16], which take spectral vectors as inputs to the model, resulting in the loss of spatial information that facilitates land cover discrimination. Particularly, the representative multilayer perceptron (MLP) [17], on the other hand, requires the training of multiple fully connected layers, which undoubtedly increases the parameter burden significantly. To reduce the number of training parameters, convolutional neural networks (CNNs), known for their local connectivity and weight sharing, are often used for feature extraction. Up to now, several CNN-based HSI classification methods have emerged, such as 1D-CNN [18], 2D-CNN [19], [20], [21], 3D-CNN [22], HybridSN which combines 2-D and 3-D convolutions [23], GNN [24], GAN [25], and transformer [26]. In particular, Hong et al. [27] proposed a universal RS foundation model for the first time called SpectralGPT, which mainly processes remote sensing images by a novel 3-D generative pretrained transformer with good results.

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In addition, Hong et al. [28] also proposed a cross-scene multisource remote sensing classification model by embedding adversarial-based domain adaptation into a neural network, which highly improves the classification of target domain scenes. However, the powerful feature extraction capability of CNN can easily lead to overfitting. Fortunately, residual structures that integrate shallow and deep features can alleviate this contradiction [29]. It is important to note that training on a large number of parameters in a deep learning model often requires the provision of a huge number of labeled samples. However, manual labeling of HSI has always been a time-consuming and laborious task. Therefore, it is still a challenging task to accomplish the training of a large number of parameters in deep learning with a limited number of samples being labeled.

In the background of limited labeled data, few-shot learning (FSL) has gradually attracted extensive attention from scholars at home and abroad again in recent years. This method is an application of meta-learning in few-shot scenarios, which is inspired by the human visual system, i.e., an object with an unknown category can be attributed to a category based on its degree of similarity to an object with a known category [30]. Accordingly, a series of FSL-based methods have been proposed and introduced into the field of HSI classification. In general, these methods first divide the training samples into support samples for use as category prototypes and query samples for comparison with the support samples, and achieve the learning of the embedding transformation function by comparing the similarity between the two samples, so that the samples to be categorized are mapped into the embedding space based on the distance metric, and then with the help of a classifier based on the distance measure, the new samples can be easily distinguished. For example, Zhang et al. [31] proposed a global prototype learning strategy in which training is learned such that each category corresponds to a vector called the global prototype representation, and the new samples are compared with the global prototype representation of each category for the purpose of categorization. The FSL method is capable of extracting the general knowledge about the training samples, also known as domain-invariant information, because this knowledge is usually independent of the distribution domain in which the samples are located. Applying this domain-invariant information as a prior knowledge to cross-domain FSL methods for new target domain data will effectively improve the classification performance in the target domain. For example, Bai et al. [32] proposed an HSI classification method that combines subspace techniques and feature transformations to achieve the feature classification in few-shot scenarios by determining the respective adaptive subspaces for different categories and by further introducing feature enhancement transformation. In [33], contrast learning was introduced into cross-domain FSL with the aim of overcoming the high interclass similarity and large intraclass variance of the samples in the metric space. However, due to its equal penalization of same-class and different-class samples, it is challenging to flexibly optimize the distribution of samples in the metric space. Li et al. [34] proposed FSCF-SSL by combining FSL and contrastive learning while setting rotated labels as a self-supervised process. Zhao et al. [35] proposed a self-motivated ensemble learning model that induces unlabeled samples to generate pseudo-labels to perform a semi-supervised self-training process. Qin et al. [36] designed FDFSL to reduce domain-specific shifts by feature decoupling via an orthogonal low-rank feature disentanglement method, which suppresses specific source domain knowledge and encourages the model to learn specific information from the target domain. Data augmentation by adding noise to target-domain labeled samples has been widely used in FSL, which may cause spectral variability. Li et al. [37] combined low-rank representations with deep learning to propose an interpretable anomaly detection network. Hong et al. [38] proposed an augmented linear mixture model for coping with spectral variability in HSI. Furthermore, it has been confirmed in [39] that the traditional one-hot encoding can make the model overconfident when updating the network parameters, leading to a generally low noise immunity of the model. In the FSL tasks, the effect of category labeling noise will be further amplified, leading to poor robustness and low robustness of the model. Therefore, it is crucial to improve the noise immunity of the FSL model.

Although many FSL methods have achieved good results, cross-domain FSL approaches still face great challenges: one is for the FSL-based HSI classification tasks, although the use of Gaussian noise to augment the training samples in the target domain has been proven to be effective and widely used, however, the overconfidence of nonlinear discriminators in sample classification tends to result in models with generally weak noise immunity. Second is the limitations of metrics space in the few-shot task, i.e., confusion between samples of different categories in the consistency metric space hampers the effective differentiation between different categories of features. The third is how to quickly extract more detailed hyperspectral features under limited labeling conditions to achieve high-precision classification of target features.

To address the above issues, this study proposes a CLSM, which combines the category label smoothing strategy with the FSL task to introduce appropriate uncertainties and encourage the discriminator to output soft probabilities, which makes the model more inclined to learn the overall pattern of the data rather than the distribution of noise, thus mitigating the negative impact of noise and improving the stability and robustness of the model. Meanwhile, this strategy also establishes partial connections between categories and reduces the standardization of features, which serves as data augmentation and sample clustering to some extent.

Furthermore, to reduce the occurrence of category misclassification so that the metric space has better category separability, this study designs an MSOM, which maximizes the intraclass similarity and minimizes the interclass similarity more flexibly by assigning different weights to the intraclass and interclass similarity scores of the sample distributions in the embedding space, respectively. This approach brings similar samples closer together while pushing dissimilar samples apart, thus optimizing the distribution of samples in the metric space and ultimately enhancing category separability. Besides, a lightweight feature extractor, LFE-MAs, skillfully

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integrating multiple attention mechanisms, has been designed to extract features from limited samples in a more efficient and comprehensive way. This feature extractor, by focusing on channels, space, and self, is able to better utilize the information in HSI, thus contributing to the enhancement of the model's classification performance. The main contributions of this study can be summarized as follows.

1) A novel FSL with label smoothing and metric space optimization (LMFSL) framework is proposed for few-shot HSI classification by combining the category label smoothing strategy with the FSL tasks, along with a further metric space optimization approach to achieve the current state-of-the-art classification performance with a limited number of labeled samples.
2) A new FSL training method is proposed, which involves a CLSM that combines labeling the category label smoothing with FSL tasks. It encourages the output of soft probabilities by reducing the confidence level of the nonlinear discriminator.
3) An MSOM is proposed, which enhances the aggregation of the query set and optimizes the distribution of embedding features of the query samples in the metric space, thus guiding the updating of prototypes for each category and enhancing the class separability of samples in the metric space.
4) In this study, a lightweight feature extraction network LFE-MAs integrating multiple attention mechanisms is proposed specifically for few-shot scenarios. The network, utilizing different attention mechanisms, focuses on both global contextual information and local information, thus enabling more effective embedded feature extraction with higher efficiency.

II. Related Work

A. Few-Shot Learning

FSL is a method in machine learning dedicated to achieving high-precision classification with a very small amount of labeled data, and is often regarded as a specific implementation of meta-learning in supervised learning. Specifically, it consists of two main phases, including a training phase and a testing phase. The training phase process can be divided into several subtasks, each of which contains a support set for forming the category prototype and a query set for comparing the similarity with the support set. The model is trained by comparing the similarity between these two sets, a process also known as episodic training, which updates the model parameters in a more efficient way [40]. The testing phase process is conducted with a limited number of labeled samples and new unlabeled samples, respectively. Furthermore, the metric-based FSL refers to category prediction of samples by training a metric space with category separability [41]. Due to its remarkable generalization performance, it has received widespread attention in the field of HSI classification.

For example, Liu et al. [42] proposed a metric-based DFSL method, which first employs a deep 3-D residual network to learn the prior knowledge from the source domain datasets. By considering the samples' embedding features, the samples of the same category are clustered near their prototypes, and then the category to which an unknown sample belongs can thus be predicted based on the distance of the individual sample from its category prototype. However, since the labeled samples of the target domain are not used to train the feature extraction module in the DFSL method, this results in inadequate extraction of information from the limited samples. Furthermore, Li et al. [43] proposed DCFSL, which alternates between source FSL and target FSL to enhance the utilization of target domain samples. As for the domain shift between source and target domains, a conditional domain adversarial strategy is introduced to align the source and target domains for extracting the domain-invariant information from the source domain so as to improve classification accuracy in the target domain. To increase the model's focus on global features and enhance the model's capability in dealing with nonlocal information, Zhang et al. [44] adopted the approach of combining graph information aggregation with domain adversarial discrimination strategies, which can effectively mitigate the negative impact of domain shift. Xi et al. [45] mapped the space-spectral features of HSI into a class-covariance metric space to optimize the decision boundaries of nonlinear classifiers, thereby enhancing the category separability of in the embedding space. Basnet et al. [46] proposed an ACDFSL method by introducing the squeeze-and-excitation (SE) attention mechanism to the DCFSL method. In order to enhance the stability of the prototype, Liu et al. [47] proposed a refined prototypical contrastive learning network named RPCL-FSL, by imposing triple constraints based on contrastive learning, self-calibration, and cross-calibration on the support set.

All the above methods show that FSL has better classification performance than deep learning for HSI classification in scenarios with limited labeled samples. However, the current FSL based on metric learning usually does not consider enough information such as intraclass similarity and interclass similarity of samples, which leads to the difficulty of resolving the confusion of samples of different categories in the embedding space, which greatly increases the difficulty of correctly classifying the samples. To address this problem, inspired by Sun et al. [48], an MSOM is designed in this study, which aims to capture the interdependence between the embedded features of the query samples in a fine-grained way, and to maximize intraclass similarity and minimize interclass similarity by assigning different weights to intraclass/interclass similarity, respectively, so as to determine more accurate category decision boundaries to reduce the likelihood of misclassification, and ultimately improve the overall classification performance of the model.

B. Label Smoothing

In traditional classification tasks, the one-hot encoding representation is often adopted, i.e., the probability of the target category is set to 1, and the probability of other nontarget categories is set to 0. However, this representation has been proved to be problematic in [39]: the model will push itself

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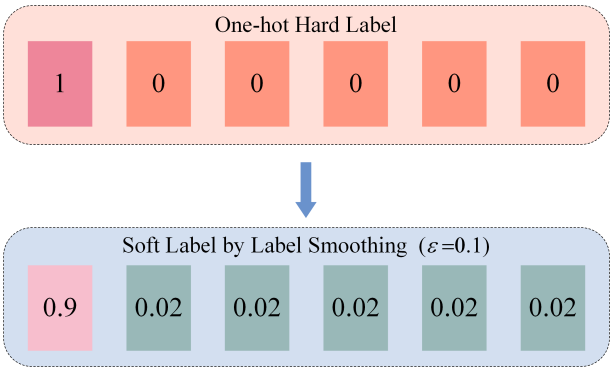


Fig. 1. Label smoothing for six categories with a smoothing parameter of 0.1.

to learn in the direction of maximizing the difference between the correct labels and the incorrect labels during the training process, so that the model will become a "black-or-white" model. This approach, especially if the amount of training data is insufficient, may fail to characterize all sample features, leading to overconfidence in the model and potentially causing overfitting. To solve this problem, Song et al. [49] proposed an unsaturated coding approach, which reduces the absolutization of category determination by decreasing the probability of a sample belonging to the target category and increasing the probability of discriminating it as a nontarget category in order to improve the noise immunity of the model. However, this way of simply changing the calculation of category attribution probabilities may result in the sum of all category probabilities no longer being equal to one, and thus violates probabilities no longer being equal to one, and thus violates the basic requirements of the Bayesian criterion. As shown in Fig. 1, the label smoothing approach normalizes all the category probabilities, i.e., the reduction in the probability values attributed to the target category is equally distributed to the nontarget category, and thus better results are expected to be achieved. For example, He et al. [50] introduced label smoothing into the field of HSI classification, and effectively improved the noise immunity of the model by combining it with a vision-transformer. In the cross-domain FSL task, data augmentation by adding Gaussian noise to the training samples of the target domain is a widely used approach to address the lack of available samples in few-shot scenarios [33], [43], [44], [45], [46], [47]. However, due to the influence of domain shift characteristics between different domains, the performance of classifiers trained on the above-expanded data can hardly be fully guaranteed in terms of noise immunity, accuracy, and stability.

In the FSL task, the introduction of a label smoothing technique to redistribute the category label probabilities helps to reduce the degree of dependence of the classifier on some specific categories, thus avoiding the model from being too arbitrary in the category determination, which effectively enhances the model's anti-noise performance. At the same time, this approach also establishes partial links between categories and reduces the degree of standardization of sample features, which to some extent serves as data augmentation and sample clustering, thus further improving the model's ability to discriminate between samples.

C. Feature Extractor for HSI Classification

The traditional HSI classification methods, such as PCA [7], LDA [8], and SVM [12], tend to perform poorly because these methods generally suffer from limitations such as shallow extracted features, data sparsity, and a disconnection between feature extraction and classification processes. In contrast, deep learning, especially CNNs [18], [19], [20], [21], [22], [23], [24], [25], exhibits significant advantages in HSI image classification by virtue of its mechanisms such as end-to-end optimization and local connection weight sharing. Inspired by visual attention, an attention mechanism was proposed and widely adopted in deep learning. It was later gradually introduced into the field of HSI classification, specifically including spatial, spectral, and self-attention mechanisms. For example, Song et al. [51] embedded a channel attention module (CAM) [52] and spatial attention module (SAM) [53] into CNN, which effectively improved the classification performance of the network. Sun et al. [54] combined the transformer structure with multihead self-attention and hybrid convolution, which not only improved the ability of capturing global information, but also strengthened the ability of mining local information. In a few-shot scenario, the sparsity of HSI is further amplified, which poses new challenges for traditional models to perform effective feature extractions. In response, Liu et al. [42] designed a deep 3-D residual network to extract the joint spatial–spectral features of HSI, however, only 3-D convolution was involved in the feature extraction process of this model, which resulted in insufficient mining of spatial information. Xi et al. [45] developed a lightweight cross-scale convolution network by improving octave convolution to achieve 2-D–3-D hybrid convolution, thus ensuring that both high and low-frequency HSI features can be considered simultaneously. Although this network model has high operational efficiency, its extraction of global information is not sufficient. Xu et al. [55] designed a deep residual hybrid attention network, which extracts HSI features through a global attention module and a local self-attention module. However, the network did not emphasize the spatial and channel information of the samples, and the local self-attention module consumes a large amount of computational resources, which results in the network operating inefficiently.

To quickly and effectively extract features from HSI under a few-shot scenario, this study designs a lightweight feature extractor LFE-MAs, incorporating the global self-attention, spatial attention, and channel attention mechanisms, resulting in a strong feature extraction capability. To further enhance the nonlinear fitting ability of LFE-MAs, the feature extractor also introduces multilevel residual connections to integrate features from different layers, which ultimately improves the discriminative ability and generalization performance of the model.

III. Proposed LMFSL for HSI Classification

As shown in Fig. 2, the framework proposed in this article, named LMFSL, primarily consists of four major components: an adjustable adaptive spectrum aligner (ASA), a lightweight feature extractor LFE-MAs, a metric space optimization module (MSOM), and a category label smoothing module (CLSM).

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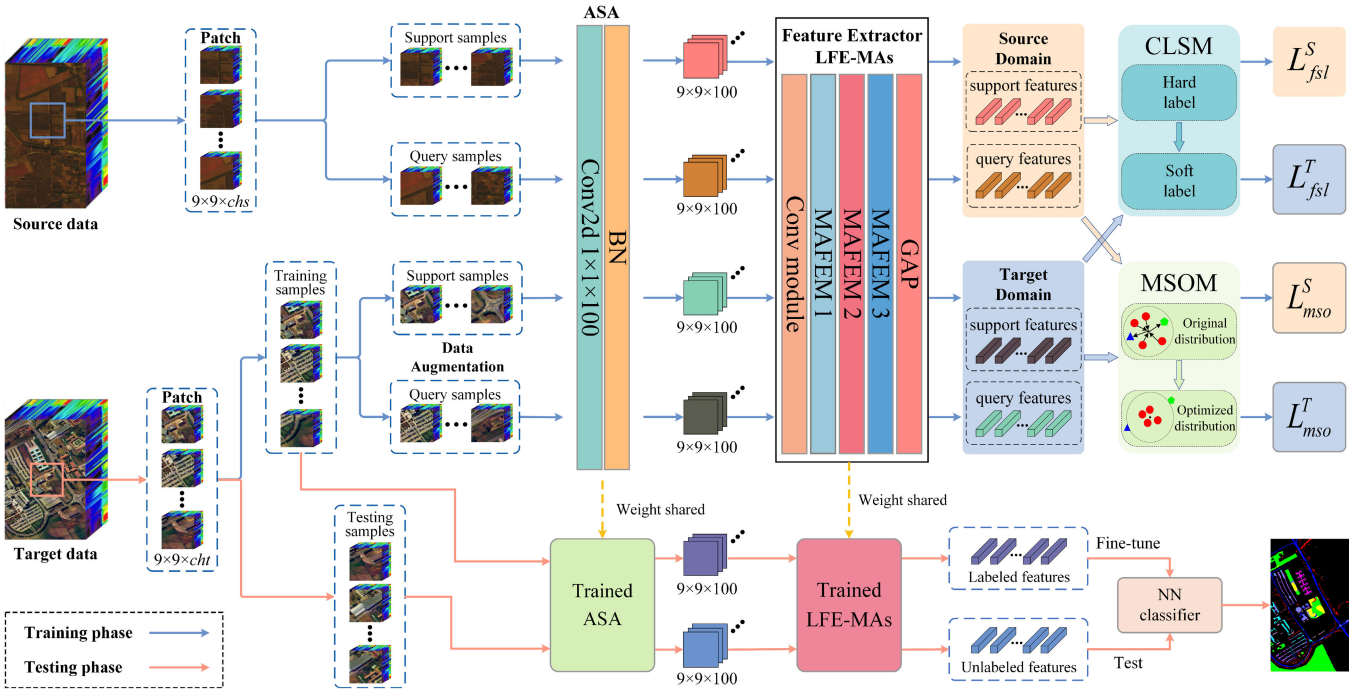


Fig. 2. Framework of the proposed LMFSL. The network mainly concludes with an ASA, a lightweight feature extractor LFE-MAs, a CLSM, and an MSOM.

With the help of a 1 × 1 2-D convolution, the ASA unifies the channel dimensions of the source and target domain data, and at the same time enhances the nonlinear fitting performance of the model. Then, the LFE-MAs are responsible for extracting the global information and local detailed features of different data domains and mapping these features into a unified embedding space based on Euclidean distances to improve the performance of the model in terms of category discriminability. Subsequently, to make the model prediction more robust, the CLSM reassigns the category label probabilities to reduce the dependence of the classifier on some specific categories, which enables the model to have better generalization performance. Moreover, the model considers the similarities and differences between the different categories during the learning process, which provides intercategory correlations and serves as a data enhancement to some extent. Besides, the MSOM works by maximizing intraclass similarity and minimizing interclass similarity, thereby reducing the distance between similar class samples and increasing the distance between dissimilar class samples. This optimizes the distribution of samples, enhances the aggregation of category prototypes, and significantly lowers the risk of misclassification by the model.

LMFSL mainly includes two stages: an alternating FSL training phase with source and target domain training samples, and a classification phase of test samples in the target domain. The training phase is performed by alternately executing FSL training in the source and target domains to parameterize the ASA and LFE-MAs so that the source and target domain data can be mapped to the embedding space with consistent scenarios. During the testing phase, the trained ASA and LFE-MAs are used to perform feature extraction on the test data, followed by classifying them using the nearest neighbor (NN) classifier. Subsequently, the classification maps are then generated so as to achieve a comprehensive evaluation of the performance of the LMFSL framework.

A. Adaptive Spectrum Aligner

In LMFSL, the HSI of the source and target domains typically differ in the spectral dimensions. To map the features from different domains into a consistent embedding metric space, an algorithm called ASA is introduced, which ensures the consistency of the cross-domain spectral dimensions by uniformly adjusting the channel numbers in the source and target domains to be the same by using a 1 × 1 2-D convolution.

For example, ASA is employed to map the source domain (e.g., Xiong'an (XA) dataset with 256 bands) and the target domain (e.g., Indian Pines (IP) dataset with 200 bands) to a consistent dimension of 100. The output of ASA can be represented as follows:

$$I' = I \times T \tag{1}$$

where $I \in \mathbb{R}^{9\times9\times\mathrm{ch}}$ represents the input 3-D cube data, $I' \in \mathbb{R}^{9\times9\times100}$ is the transformed 3-D cube data, and $T \in \mathbb{R}^{\mathrm{ch}\times100}$ denotes the transformation matrix, and ch × 100 refers to the trainable transformation parameters.

B. Lightweight Feature Extractor With Multiple Attention

To extract from variable HSI the highly discriminable embedding features that can be mapped to a consistent embedding metric space, an advanced lightweight feature extractor LFE-MAs, is designed in this study, and its structure is shown in Fig. 3. The LFE-MAs consists of an initial convolution block and three multiple attention feature extraction (MAFE) blocks. By incorporating multilevel residual connections, this

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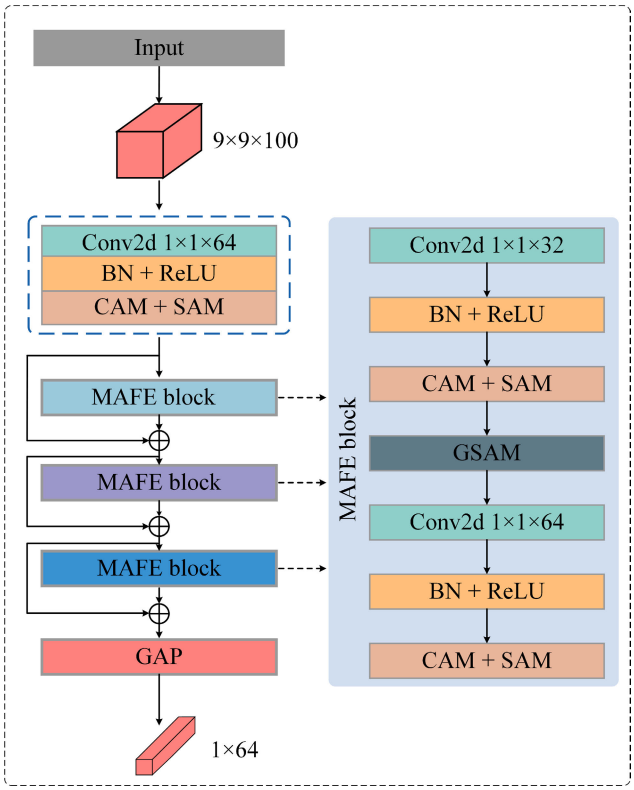


Fig. 3. Architecture of the proposed LFE-MAs.

extractor effectively integrates the spatial–spectral information in both shallow and deep layers. The convolution module processes the HSI using a 1×1 2-D convolution, an operation that not only adjusts the spectral dimensions of the data to fit the subsequent MAFE blocks, but also enhances the nonlinear fitting ability of the model. In addition, the integrated ReLU activation function can also enhance the model's nonlinear fitting ability, while the BN layer helps to accelerate the convergence speed of the network. The MAFE block incorporates channel, spatial, and global self-attention to focus on more discriminative features in HSI by paying attention to channel, spatial, and its own characteristics, so as to efficiently extract the local and global information from HSI, thus significantly improving the feature extraction capability of the model.

1) Channel Attention Module (CAM): CAM [51] aims to capture the importance of each channel dimension in the feature map and assign different weights accordingly. As shown in Fig. 4(a), the initial feature map undergoes global maximum pooling (GMP) and global average pooling (GAP) operations, respectively, to generate two different sets of feature maps. These two sets of feature maps are then fed into MLP for weight sharing. The output feature maps from the MLP are further summed and normalized through a Sigmoid function to obtain the channel attention weights M_{C}. Subsequently, M_{C} is elementwise broadcast multiplied with the original feature map F to output the final feature map F'. All the above computational processes can be represented as follows:

$$M_{\mathrm{C}}(F) = \sigma(\mathrm{MLP}(\mathrm{GMP}(F)) + \mathrm{MLP}(\mathrm{GAP}(F))) \tag{2}$$

$$F' = F \odot M_{\mathrm{C}}(F) \tag{3}$$

where σ refers to sigmoid function normalizing the weight values to $[0, 1]$, and $\odot$ is the elementwise broadcast multiplication operation.

2) Spatial Attention Module: As shown in Fig. 4(b), the SAM [52] focuses on the spatial information of the feature map so as to accurately capture the importance of each spatial region in the feature map and assign weights accordingly. The input feature map F first undergoes GMP and GAP operations along its spectral dimension to produce two feature maps which contain different spatial information. Subsequently, these two feature maps are merged and subjected to a 7×7 2-D convolution to generate a single-channel feature map F_{conv}. Then, this feature map is normalized by a Sigmoid function to the spatial attention weights M_{S}. Eventually, the output feature map F' is obtained by performing an elementwise broadcast multiplication operation of M_{S} with the initial feature map F. The whole calculation process can be represented as follows:

$$M_{\mathrm{S}}(F) = \sigma(\mathrm{Conv2d}(\mathrm{GMP}(F), \mathrm{GAP}(F))) \tag{4}$$

$$F' = F \odot M_{\mathrm{S}}(F). \tag{5}$$

3) Global Self-Attention Module (GSAM): As shown in Fig. 4(c), GSAM [55] aims to finely model the global feature representation and the autocorrelation between pixels. By generating a similarity mask, the module not only enhances the relevant information, but also effectively suppresses irrelevant information. To be more specific, the input feature map F is first subjected to a GAP operation to obtain the global feature representation F_{avg}, which is subsequently reshaped to a dimension of $1 \times C$ and subjected to $L2$ regularization. Meanwhile, the original feature map F is reshaped to a dimension of $HW \times C$ and also subjected to $L2$ regularization. The two feature maps after $L2$ regularization are then matrix multiplied and processed by the softmax function to produce the normalized similarity mask M. What follows is M being reshaped to a global self-attention weight M_{self}. This weight is then elementwise broadcast multiplied with the original feature map F to generate the feature map $\hat{F}$. Finally, the output feature F' is obtained by connecting F to $\hat{F}$ through the residual connection. The whole calculation process above can be represented as follows:

$$F_{\mathrm{avg}} = \mathrm{GAP}(F) \tag{6}$$

$$M = \mathrm{softmax}\left(\left(\frac{F}{\|F\|_2}\right)\left(\frac{F_{\mathrm{avg}}}{\|F_{\mathrm{avg}}\|_2}\right)^{\mathrm{T}}\right) \tag{7}$$

$$F' = \mathrm{sum}(F, (F \odot M_{\mathrm{self}})). \tag{8}$$

C. CLSM for FSL

In LMFSL, the source domain HSI dataset D_{S} contains C_{S} classes, while the target domain HSI dataset D_{T} consists of C_{T} classes. Depending on whether it contains the labeled data or not, D_{T} is further divided into a few-shot dataset D_{f} for training and a dataset D_t for classification testing, where $D_{\mathrm{T}} = D_{\mathrm{f}} \cup D_t$. Due to the limited number of samples in D_{f}, which is insufficient to perform the FSL training process, a data augmentation approach is performed on D_{f}

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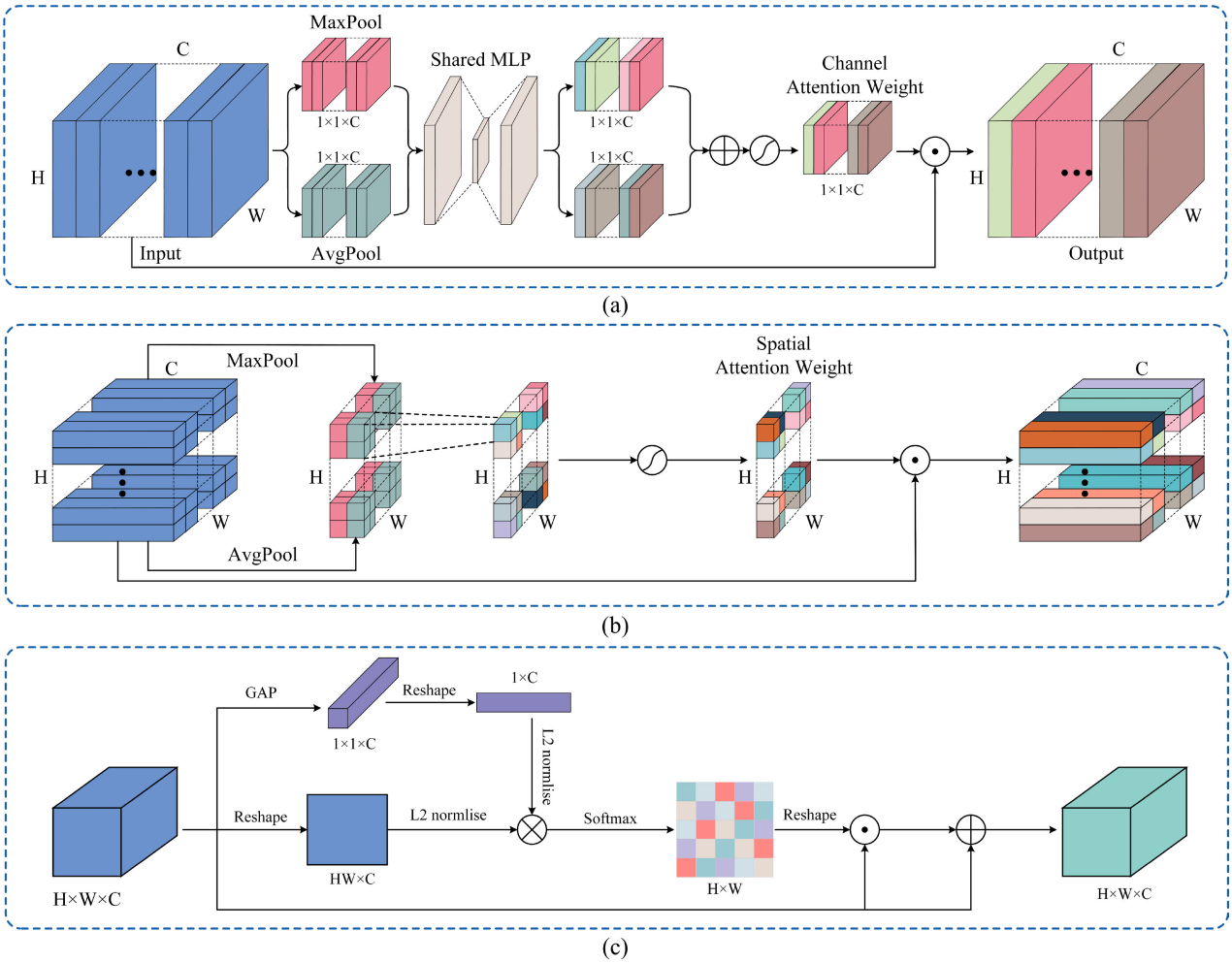


Fig. 4. Flowchart of multiple attention mechanisms. (a) CAM. (b) SAM. (c) GSAM.

by adding random Gaussian noise to it. The training process employs a strategy of alternating FSL between the source domain and the target domain. This strategy aims to learn domain-invariant information from the source domain as well as some of the domain-related information that is valid for the sample classification in the target domain. Both types of information are further mapped to the consistency metric space, thus reducing the negative impacts of domain shift and enhancing the model feature extraction capability. Specifically, in the source domain FSL, N_S categories are first randomly sampled from the C_S categories in the source domain dataset D_S, and K_s samples are selected from each category to form the support set $S_s = \{(x_i, y_i)\}_{i=1}^{N_S \times K_s}$. Where $x_i \in \mathbb{R}^{H \times W \times \text{chs}}$ represents the ith sample in the source domain support set, chs represents the number of bands of the source domain HSI, and y_i is the corresponding label of x_i. Meanwhile, K_q samples are selected from each of the rest of the same N_S categories to form the query set $Q_S = \{(x_j, y_j)\}_{j=1}^{N_S \times K_q}$. Where (x_j, y_j) represents the jth sample in the query set and its corresponding label. Subsequently, the support set and query set are reorganized into a $K_q - \text{way} - K_s - \text{shot}$ episodic task to perform the source domain FSL. Similarly, in the target domain FSL, random sampling from the augmented few-shot dataset D_f is performed to form the support set $S_T = \{(x_i, y_i)\}_{i=1}^{N_T \times K_s}$ and the query set $Q_T = \{(x_j, y_j)\}_{j=1}^{N_T \times K_q}$ to perform the target domain FSL. It is noteworthy that the FSL in the two domains are conducted alternatively without interfering with each other. The category distribution for a query sample x_j in the query set Q_S is depicted as follows:

$$P\left(y_j = k \middle| x_j \in Q_S\right) = \frac{\exp\left(-d\left(f_\varphi\left(x_j\right), f_k\right)\right)}{\sum_{k=1}^{N_T} \exp\left(-d\left(f_\varphi\left(x_j\right), f_k\right)\right)} \tag{9}$$

where $d(\cdot, \cdot)$ refers to the Euclidean distance function, f_k is the embedding feature of the kth class in the support set, N_T is the number of classes in each episode, and f_φ refers to the ASA and LFE-MAs with optimization parameter φ. According to the negative logarithmic probability x of the query sample and its true label k, the classification loss of the source domain episodic training can be formulated

$$L_{\text{fsl}}^{S} = E_{Q_S, S_S}\left[-\sum_{(x,y)\in Q_S} \log p_\varphi(y = k|x)\right]. \tag{10}$$

Analogously, the classification loss of the target domain episodic training can be formulated below

$$L_{\text{fsl}}^{T} = E_{Q_T, S_T}\left[-\sum_{(x,y)\in Q_T} \log p_\varphi(y = k|x)\right]. \tag{11}$$

To prevent the prediction of the model from being overconfident while enhancing its resistance to noise, label smoothing

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is performed on the category labels. Based on the total number of categories N and the label smoothing parameter ε, the probability of the correct category label is reduced by a portion and equally distributed to other categories. Such smoothing process can be defined as follows:

$$\hat{y}_i = \begin{cases} 1-\varepsilon+\dfrac{\varepsilon}{N}, & i=k \\ \dfrac{\varepsilon}{N}, & i\neq k. \end{cases} \tag{12}$$

Therefore, in the FSL task that incorporates label smoothing, the classification loss for episodic training in the source domain can be represented as follows:

$$L_{\text{fsl}}^{S} = E_{Q_S,S_S}\left[-\sum_{(x,y)\in Q_S}\sum_{k=1}^{N_S}\hat{y}_k \log p_\varphi(y=k|x)\right]. \tag{13}$$

Similarly, in the FSL task that incorporates label smoothing, the classification loss for episodic training in the target domain can be expressed as follows:

$$L_{\text{fsl}}^{T} = E_{Q_T,S_T}\left[-\sum_{(x,y)\in Q_T}\sum_{k=1}^{N_T}\hat{y}_k \log p_\varphi(y=k|x)\right]. \tag{14}$$

D. Metric Space Optimization Module

When metric-based FSL maps the source and target domain samples to the consistency metric space to address the domain shift that exists between the source and target domains, one of the most prominent characteristics is that the distance between the distributions of samples of the same category is too large while the distance between the distributions of samples of different categories is too small, which can easily lead to misclassification of the sample categories. To overcome this, a module of MSOM is proposed to optimize the spatial distribution of samples in the metric space, as illustrated in Fig. 5, This module works by drawing samples of the same category closer together and pushing samples of different categories farther apart, thereby increasing the aggregation degree of category prototypes and enhancing the stability and robustness of the model. First, it is necessary to assess the similarity between different samples. The intraclass similarity score s_p^i for the ith sample among K same category samples, and the interclass similarity score s_n^j for the jth sample among L different category samples can be defined, respectively, as follows:

$$s_p^i = \frac{1}{K-1}\sum_{k=1,k\neq i}^{K}\frac{f_i\cdot f_k}{\|f_i\|_2\times\|f_k\|_2} \tag{15}$$

$$s_n^j = \frac{1}{L-1}\sum_{l=1,l\neq j}^{L}\frac{f_i\cdot f_l}{\|f_i\|_2\times\|f_l\|_2} \tag{16}$$

where f_i and f_j are the extracted sample features, f_k is the sample feature of the same category different from f_i, and f_l is the sample feature of different category different from f_j. The degree of aggregation of category prototypes can be increased by maximizing the intraclass similarity score s_p while minimizing the interclass similarity score s_n. In the traditional methods, the main focus is on improving and optimizing for $(s_p - s_n)$, and the immediate consequence is the equal strength of the penalties for s_p and s_n, which leads to a certain lack of flexibility in gradient updating. Therefore, two independent weight factors α_p and α_n are defined in MSOM, which are linearly correlated with s_p and s_n, respectively, transforming the optimization problem from $(s_p - s_n)$ to $(\alpha_p s_p - \alpha_n s_n)$, which makes the network update more flexible. Besides, α_p^i and α_n^j can be defined, respectively, as follows:

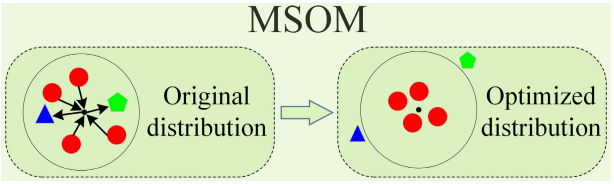


Fig. 5. Example plot of optimization using MSOM for three categories in metric space.

$$\alpha_p^i = \max\{1+m-s_p^i, 0\} \tag{17}$$

$$\alpha_n^j = \max\{s_n^j+m, 0\} \tag{18}$$

where m is a relaxation factor for controlling the radiality of the decision boundary in the metric space. Defining γ as the maximum scale factor for determining each similarity score, the loss function of the proposed MSOM can then be defined as

$$L_{\text{mso}} = \log\left[1+\sum_{j=1}^{L}\exp\left(\gamma\alpha_n^j\left(s_n^j-m\right)\right)\times\sum_{i=1}^{K}-\gamma\alpha_p^i\left(s_p^i-m+1\right)\right]. \tag{19}$$

IV. Experiment

To verify the effectiveness of LMFSL, this study carries out a comparison of the classification performance and computational complexity of LMFSL with four non-FSL methods and five FSL methods. Meanwhile, a series of ablation experiments are carried out in this article to evaluate the contribution of each proposed module. In addition, further analytical experiments are conducted to ensure the optimal parameter settings for LMFSL.

The programming and implementation of the experiments are done in the PyCharm development environment with integrated Python 3.8 and PyTorch 2.0, running on a computing platform with a 13th generation Intel Core i9-13900KF processor, an NVidia GeForce RTX 4080 graphics card, and 64 GB of memory.

A. Experimental Datasets

To comprehensively evaluate the effectiveness of the proposed LMFSL framework in cross-domain FSL, five highly representative public HSI datasets are selected for the experiments in this study, which are XA, University of Pavia (UP), Pavia Center (PC), IP, and Salinas (SA). Considering that the source domain data needs to have rich category information for effective cross-domain migration, the XA dataset, for its rich category variety, is chosen as the source domain, while the other datasets are used as the target domains for validation.

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TABLE I
LAND COVER CLASSES AND SAMPLE NUMBERS IN THE XA DATASET

Class No.	Class Name	Samples
1	Acer N	225647
2	Willow	180766
3	Elm	15353
4	Paddy	452144
5	Sophora J	475591
6	Fraxinus C	169342
7	Golden R T	23304
8	Water	165647
9	Bare G	38409
10	Acacia	193830
11	Stubble	5612
12	Corn	59165
13	Pyrus	1026513
14	Poplar	7151
15	Soybean	91072
16	Vegetable F	29148
17	Sparse F	1496
18	Peach	421790
19	Grass	65514
20	Building	29616
	Total	3,677,110

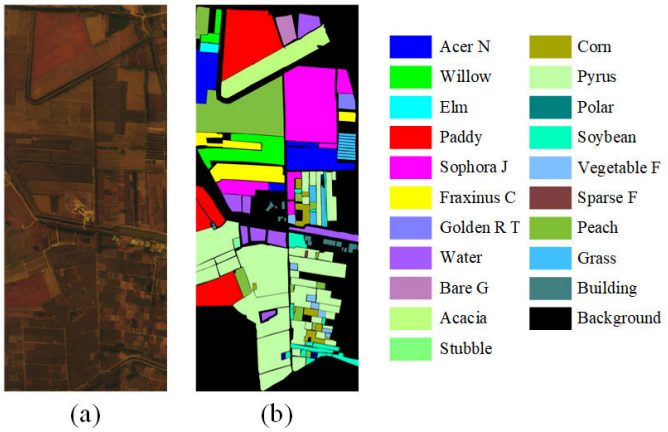


Fig. 6. XA dataset. (a) False-color image. (b) Ground-truth map.

1) XA Dataset: The XA dataset consists of the HSI of an agricultural area located in the XA New Area, Hebei Province, China, acquired by using a newly designed airborne hyperspectral sensor accompanied by synchronous ground survey experiments. Fig. 6 shows the false-color image of this dataset along with the corresponding ground-truth map. This dataset provides 256 spectral bands ranging from 400 to 1000 nm and has a spatial resolution of 1750 × 1580 pixels, with a spatial size of 0.5 m per pixel. The dataset comprises 20 classes, with detailed class information provided in Table I. A total of 20 land cover types are included in this dataset, and their detailed category information is depicted in Table I.

2) UP Dataset: The UP dataset was acquired by Germany's Reflective Optics Spectrographic Imaging System (ROSIS) at UP, Italy, in 2003, and the false-color image of this dataset, together with the corresponding ground-truth map, is shown in Fig. 7. The dataset covers 103 spectral bands within the 430–860 nm spectral range and has spatial category information is depicted in Table I. dimensions of 610 × 340 pixels, with a spatial resolution of 1.3 m per pixel. The total number of valid pixels used for classification is 42 776, and these pixels are further classified into nine categories, and the detailed category information is provided in Table II.

TABLE II
LAND COVER CLASSES AND SAMPLE NUMBERS IN THE UP DATASET

Class No.	Class Name	Training	Testing
1	Asphalt	5	6628
2	Meadows	5	18644
3	Gravel	5	2094
4	Trees	5	30659
5	Sheets	5	1340
6	Bare soil	5	5024
7	Bitumen	5	1325
8	Bricks	5	3677
9	Shadow	5	942
	Total	45	42,731

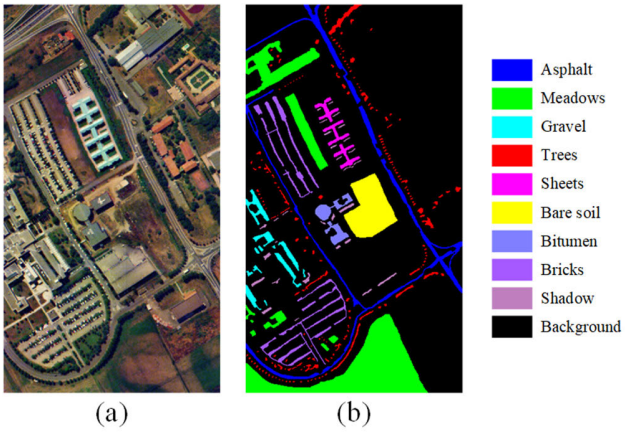


Fig. 7. UP dataset. (a) False-color image. (b) Ground-truth map.

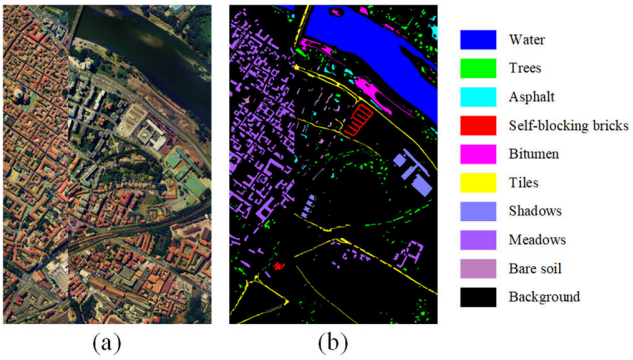


Fig. 8. PC dataset. (a) False-color image. (b) Ground-truth map.

3) PC Dataset: The PC dataset was acquired by the ROSIS sensor in the Pavia region of northern Italy. Fig. 8 displays the false-color image of this dataset along with its corresponding ground-truth map. The dataset encompasses 102 spectral bands, with a wavelength range from 430 to 860 nm, and has a spatial resolution of 1906 × 715 pixels, with a spatial resolution of 1.3 m per pixel. The total number of valid pixels used for classification is 148 197, and these pixels are classified into nine categories, which are described in Table III.

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TABLE III
LAND COVER CLASSES AND SAMPLE NUMBERS IN THE PC DATASET

Class No.	Class Name	Training	Testing
1	Water	5	65971
2	Trees	5	7598
3	Asphalt	5	3090
4	Self-Blocking Bricks	5	2685
5	Bitumen	5	6584
6	Tiles	5	9248
7	Shadows	5	7287
8	Meadows	5	42826
9	Bare Soil	5	2863
	Total	45	148,152

TABLE IV
LAND COVER CLASSES AND SAMPLE NUMBERS IN THE IP DATASET

Class No.	Class Name	Training	Testing
1	Alfalfa	5	41
2	Corn-notill	5	1423
3	Corn-mintill	5	825
4	Corn	5	232
5	Grass-pasture	5	478
6	Grass-trees	5	725
7	Grass-pasture-mowed	5	23
8	Hay-windrowed	5	473
9	Oats	5	15
10	Soybean-notill	5	967
11	Soybean-mintill	5	2450
12	Soybean-clean	5	588
13	Wheat	5	200
14	Woods	5	1260
15	Buildings-Grass-Trees-Drives	5	381
16	Stone-Steel-Towers	5	88
	Total	80	10,169

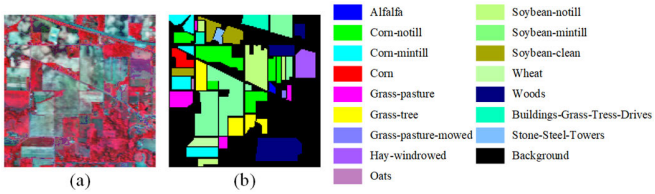


Fig. 9. IP dataset. (a) False-color image. (b) Ground-truth map.

4) IP Dataset: The IP dataset was acquired in 1992 by an Airborne Visible Infrared Imaging Spectrometer (AVIRIS) targeting the IP area. The false-color image and the corresponding ground-truth map are shown in Fig. 9. This dataset consists of a 145 × 145 pixels image with a spatial resolution of 20 m, covering 220 spectral bands. After removing 20 bands affected by water vapor absorption, 200 valid bands are finally retained. Of these, 10 249 pixels were labeled for the subsequent classification and analysis. The dataset covers 16 categories, and the specific category attributes are detailed in Table IV.

5) SA Dataset: The SA dataset was captured by the AVIRIS sensor in the SA Valley in California, USA, with its false-color image and the corresponding ground-truth map shown in Fig. 10. The dataset has an image dimension of 512 × 217 pixels and a spatial resolution of 3.7 m. This dataset

TABLE V
LAND COVER CLASSES AND SAMPLE NUMBERS IN THE SA DATASET

Class No.	Class Name	Training	Testing
1	Brocoli_green_weeds_1	5	2004
2	Brocoli_green_weeds_2	5	3721
3	Fallow	5	1971
4	Fallow_rough_plow	5	1389
5	Fallow_smooth	5	2673
6	Stubble	5	3954
7	Celery	5	3574
8	Grape_untrained	5	11266
9	Soil_vinyard_develop	5	6198
10	Corn_senesced_green_weeds	5	3273
11	Lettuce_romaine_4wk	5	1063
12	Lettuce_romaine_5wk	5	1922
13	Lettuce_romaine_6wk	5	911
14	Lettuce_romaine_7wk	5	1065
15	Vinyard_untrained	5	7263
16	Vinyard_vertical_trellis	5	1802
	Total	80	54,049

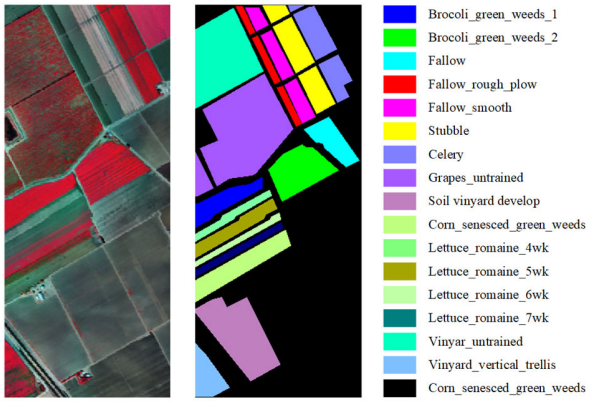


Fig. 10. PC dataset. (a) False-color image. (b) Ground-truth map.

initially contained 224 spectral bands. After excluding the bands absorbed by water vapor, the final dataset effectively covers 204 bands within the 400–2500 nm wavelength range. And there is a total of 54 129 pixels available for the subsequent classification tasks. All these pixels are further subdivided into 16 categories, with the specific information described in Table V.

B. Experimental Settings

For a fair performance evaluation, the experimental configurations of this study are set up as follows. The input size of HSI cubes of the proposed LMFSL framework is fixed as 9 × 9 × ch. Adam is chosen as the optimization algorithm, with a learning rate set at 0.001. In the source domain data, 200 labeled samples per class are randomly selected from each category for the FSL, while in the target dataset, only five labeled samples are used for training in each category and the rest of the samples are used for testing. To fulfill the requirement of FSL in the target domain, data enhancement is done here by adding Gaussian noise. A total of 10 000 episodes

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TABLE VI
CLASSIFICATION RESULT (%) ON THE UP DATASET (FIVE LABELED SAMPLES EACH CATEGORY)

Class	KNN	SVM	HybridSN	SSFTT	DFSL	DCFSL	Gia-CFSL	CMFSL	ACDFSL	RPCL-FSL	LMFSL
1	65.91±3.66	66.52±9.28	52.33±13.92	69.37±7.84	62.85±13.66	78.92±7.05	71.47±14.17	73.83±10.78	80.48±7.86	82.10±7.88	**88.52±6.46**
2	53.46±5.96	57.70±10.50	61.70±15.01	85.88±8.64	76.38±10.62	84.43±10.43	90.08±5.62	89.03±3.74	88.34±5.72	81.85±8.50	**91.29±5.28**
3	48.58±9.79	53.23±13.20	73.93±13.45	**79.75±6.31**	58.21±13.83	62.47±13.40	64.20±16.18	58.63±13.01	67.21±10.75	61.56±11.63	72.20±10.53
4	84.41±13.34	77.45±11.55	58.29±11.40	83.43±4.73	87.13±5.63	94.02±4.57	90.68±3.75	92.22±2.87	**94.82±2.45**	94.11±2.98	92.39±2.76
5	98.49±0.79	92.15±10.74	99.66±0.39	**100.00±0.00**	99.34±0.61	99.27±0.89	98.43±1.73	97.25±2.01	99.09±0.99	98.09±1.79	99.23±0.62
6	53.76±11.36	54.24±4.26	76.31±11.64	**89.67±5.56**	72.74±9.35	77.93±12.27	76.68±9.93	81.74±9.26	75.78±11.68	75.82±14.60	79.33±17.21
7	77.52±11.16	81.17±9.67	96.60±3.95	**98.26±2.63**	77.27±5.66	76.14±10.71	85.20±6.63	74.80±8.07	81.68±8.89	80.83±5.43	91.99±6.35
8	60.23±8.10	65.86±12.31	62.04±14.86	56.39±12.74	59.17±11.96	64.04±11.06	71.91±9.45	72.53±10.80	66.59±14.46	74.08±14.60	**84.16±9.69**
9	**99.80±0.11**	98.17±1.16	50.07±6.69	86.32±5.06	89.31±5.33	95.15±7.09	92.81±7.77	97.29±2.10	95.46±6.40	96.75±5.56	97.10±3.49
OA	61.17±2.03	63.25±3.96	64.37±8.18	81.59±3.05	73.29±4.30	81.11±4.73	83.00±3.03	83.13±1.50	83.49±2.30	81.20±2.95	**88.38±2.50**
AA	71.35±1.96	71.83±0.90	70.10±4.26	83.23±1.15	75.82±2.34	81.37±2.80	82.38±2.24	81.92±1.17	83.27±1.22	82.80±1.59	**88.47±2.96**
Kappa	52.13±2.15	54.21±3.84	56.29±8.44	76.24±3.49	66.10±4.73	75.67±5.45	77.82±3.71	78.03±1.87	78.49±2.81	75.81±3.42	**84.73±3.35**

TABLE VII
CLASSIFICATION RESULT (%) ON THE PC DATASET (FIVE LABELED SAMPLES EACH CATEGORY)

Class	KNN	SVM	HybridSN	SSFTT	DFSL	DCFSL	Gia-CFSL	CMFSL	ACDFSL	RPCL-FSL	LMFSL
1	98.98±0.29	97.77±1.16	95.21±3.73	99.74±0.34	98.85±0.63	**99.91±0.15**	99.71±0.26	99.63±0.39	99.92±0.09	99.48±0.94	99.50±0.28
2	81.97±7.78	79.54±8.12	69.77±5.83	85.81±1.63	89.95±3.17	91.76±4.43	**92.68±3.74**	90.60±3.96	91.31±4.01	91.45±3.95	92.34±3.21
3	71.96±11.23	78.39±7.97	63.07±7.11	92.83±2.66	78.89±9.39	93.16±3.89	90.91±5.66	**93.91±3.70**	95.01±3.49	91.28±6.98	89.47±5.99
4	68.60±8.49	65.60±5.70	83.55±11.69	**97.79±2.25**	83.76±7.84	92.96±8.27	93.12±6.25	95.65±6.37	94.12±7.05	93.03±7.61	92.54±6.69
5	65.77±14.54	77.08±9.18	68.09±7.80	67.94±2.63	78.17±9.39	90.99±5.95	88.93±6.46	88.21±6.99	90.12±5.81	90.02±5.83	**91.55±5.00**
6	91.73±6.02	88.70±7.51	46.64±10.03	97.23±1.52	80.36±7.12	95.80±3.05	95.40±2.86	93.60±5.93	96.15±2.20	96.28±2.38	**98.55±1.50**
7	73.05±5.31	76.18±10.32	85.00±1.28	**97.09±0.62**	79.52±3.53	85.31±2.85	85.46±1.39	85.11±5.49	86.28±2.84	85.18±3.10	86.92±5.14
8	85.15±12.15	83.63±12.05	71.14±11.00	96.05±1.42	92.92±4.04	98.75±0.96	98.00±1.23	97.21±1.51	98.80±1.03	98.48±1.34	**99.04±0.74**
9	99.33±1.17	99.62±0.52	69.03±7.85	92.96±2.48	89.20±4.15	97.83±1.35	95.95±2.47	**99.80±0.27**	96.90±1.93	99.05±0.80	95.99±2.74
OA	89.80±3.22	89.25±3.55	80.82±3.21	95.95±0.44	92.78±1.49	97.48±0.52	97.03±0.56	96.68±0.89	97.55±0.55	97.16±0.70	**97.57±0.48**
AA	81.84±1.75	82.95±2.28	72.39±1.99	91.94±0.44	85.74±2.74	94.05±0.88	93.35±0.96	93.75±1.41	**94.29±0.52**	93.81±1.08	93.99±1.26
Kappa	85.80±4.26	85.14±4.64	73.87±4.16	94.29±0.61	89.88±2.05	96.44±0.72	95.81±0.79	95.32±1.24	96.53±0.77	95.99±0.98	**96.56±0.67**

of training are performed during the alternate execution of source and target domain FSL. In each episode, the number of support sets is set to 1, and the number of query sets is set to be consistent with the number of categories contained in the data domains. For instance, in the source domain dataset XA, this number is set to 20, while in the target domain datasets UP and PC, it is set to 9, and in IP and SA, it is set to 16.

In addition, the selected benchmark methods, including DFSL, DCFSL, Gia-CFSL, CMFSL, ACDFSL, and RPCL-FSL follow the same settings to ensure a fair comparison. In terms of model architecture, the number of MAFE blocks in LFE-MAs is set to 3. Other hyper-parameters include a relaxation factor m being set to 0.4, a maximum scale factor γ being set to 80, and a label smoothing factor ε set of 0.1. It is worth noting that the hyper-parameters of all comparison methods, such as the learning rate, are configured according to the optimal settings provided in the relevant literatures. To comprehensively assess the performance advantages and disadvantages of the above comparison methods, three evaluation metrics, overall accuracy (OA), average accuracy (AA), and Kappa Coefficient (Kappa), are used in the experiments for the classification performance evaluations.

C. Comparison With Different Methods

To verify the effectiveness of the proposed LMFSL in HSI classification with limited samples, a series of rigorous comparative experiments are conducted in this article. The experiments include the performance comparisons with four non-FSL algorithms (KNN, SVM, HybridSN, and SSFTT) and five state-of-the-art FSL algorithms (DFSL, DCFSL, Gia-CFSL, CMFSL, ACDFSL, and RPCL-FSL). Tables VI–X show the classification performance of LMFSL compared to different methods on four test datasets, where in all methods, five labeled samples are randomly selected for training in each category separately and the remaining samples are used as the tests. For statistical significance and reproducibility, all experiments are performed ten times and the visualized classification results are shown in Figs. 11–14. In addition, to further analyze and evaluate the category separability of different FSL methods, this study utilizes the t-SNE algorithm to perform the 2-D visualization of the feature vectors extracted by each method. The results are illustrated in Figs. 15 and 16.

1) Non-FSL Methods: Four traditional supervised methods are selected in this study, including KNN, SVM HybridSN, and SSFTT. Both KNN and SVM algorithms are classical machine learning classification algorithms. The KNN algorithm performs classification by calculating Euclidean distances between the test samples and training samples. The SVM algorithm maps the high-dimensional features to the low-dimensional ones by using the kernel functions to find the optimal hyperplane for classification. However, SVM focuses mainly on spectral characteristics and tends to ignore the spatial information of HSI. HybridSN enhances classification performance by effectively extracting spatial–spectral features through the hybrid use of 3-D and 2-D convolutions, while SSFTT combines the transformer structure on the basis of the hybrid convolution to capture both the spatial and spectral

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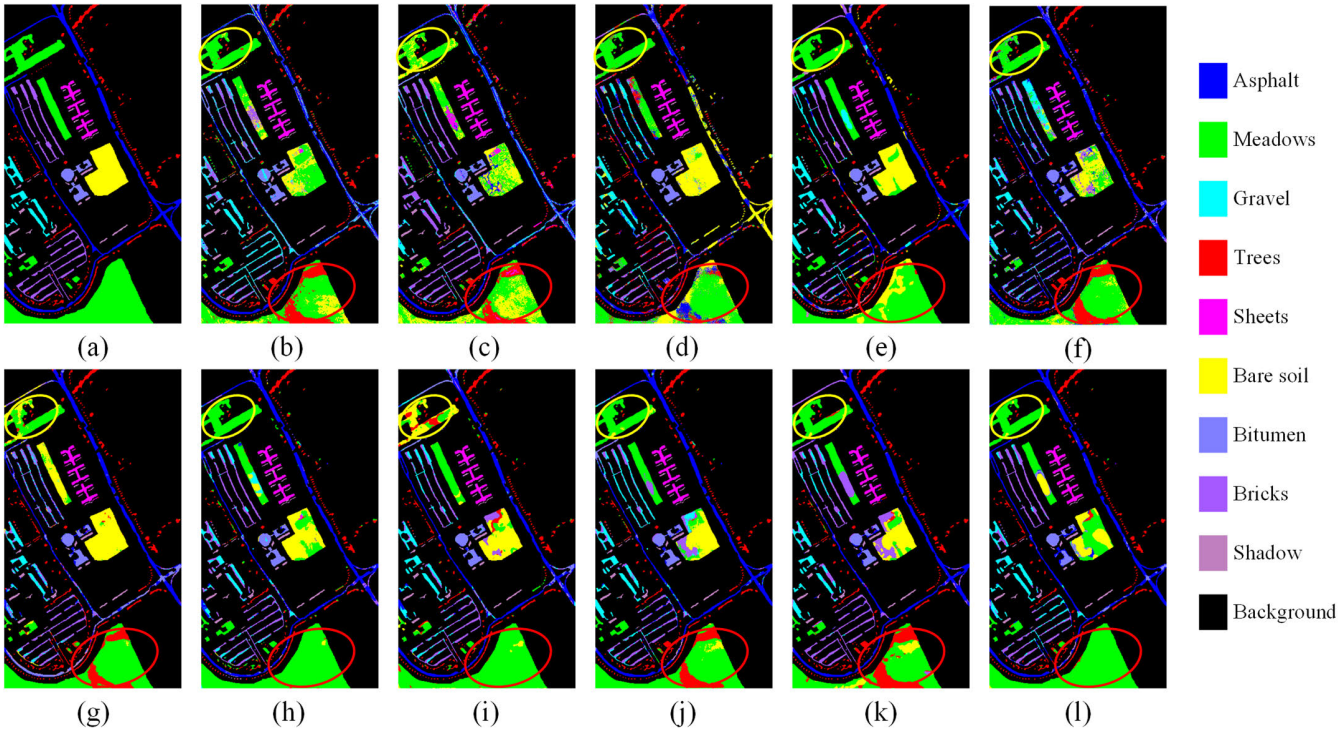


Fig. 11. Classification map on the UP dataset. (a) Ground truth. Classification maps via various algorithms for the UP dataset. (b) KNN, (c) SVM, (d) HybridSN, (e) SSFTT, (f) DFSL, (g) DCFSL, (h) Gia-CFSL, (i) CMFSL, (j) ACDFSL, (k) RPCL-FSL, and (l) LMFSL.

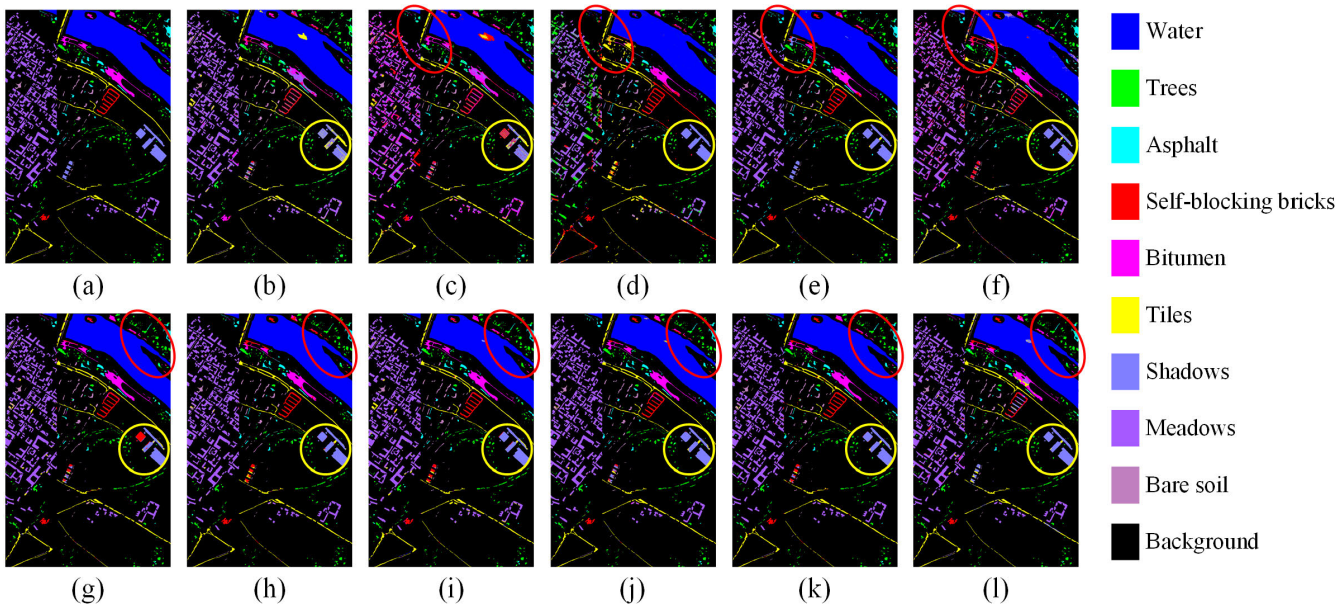


Fig. 12. Classification map on the PC dataset. (a) Ground truth. Classification maps via various algorithms for the PC dataset. (b) KNN, (c) SVM, (d) HybridSN, (e) SSFTT, (f) DFSL, (g) DCFSL, (h) Gia-CFSL, (i) CMFSL, (j) ACDFSL, (k) RPCL-FSL, and (l) LMFSL.

features more comprehensively. During the experiments, K is set to 1 for the KNN method, and a linear kernel is chosen for the SVM. In addition, to evaluate the performance of these non-FSL methods in a few-shot scenario, five labeled samples for each category are selected for training and the rest samples are used as the test set.

2) FSL Methods: Five advanced FSL methods are adopted for this study: DFSL, DCFSL, Gia-CFSL, CMFSL ACDFSL, and RPCL-FSL. DFSL first extracts the general domain-invariant features from the source domain data that are useful for the classification of the target domain, and subsequently, it constructs a metric space based on the Euclidean distance so as to facilitate the proximity of similar samples in that space. Then, to enhance the classification boundaries, this method uses a NN classifier for the classification of unknown samples. Based on DFSL, DCFSL makes the metric space more category-differentiable by alternating the training of source and target domains. In addition, this method further introduces a domain discrimination module based on adversarial training, which is used to mitigate the

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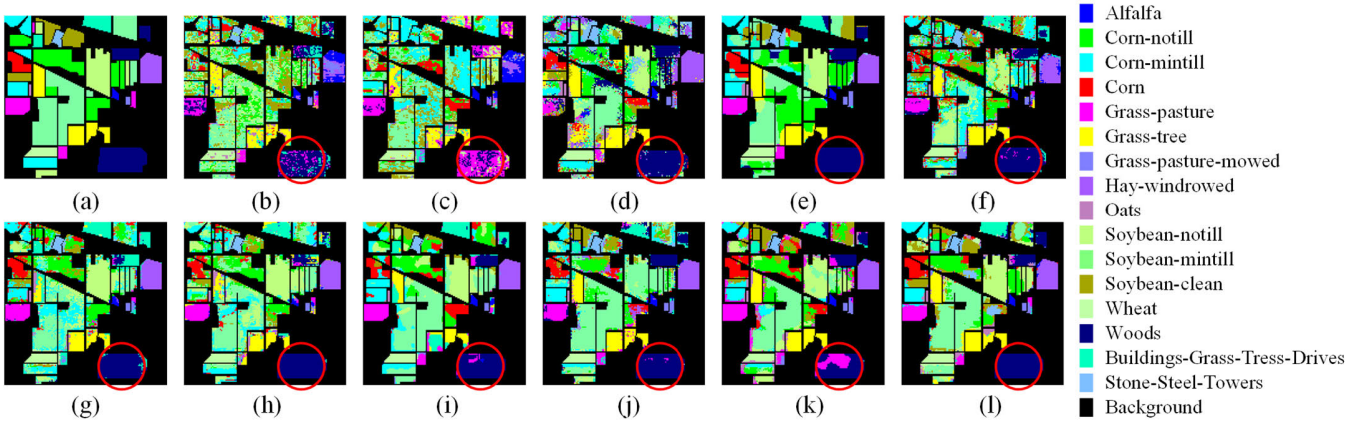


Fig. 13. Classification map on the IP dataset. (a) Ground truth. Classification maps via various algorithms for the IP dataset. (b) KNN, (c) SVM, (d) HybridSN, (e) SSFTT, (f) DFSL, (g) DCFSL, (h) Gia-CFSL, (i) CMFSL, (j) ACDFSL, (k) RPCL-FSL, and (l) LMFSL.

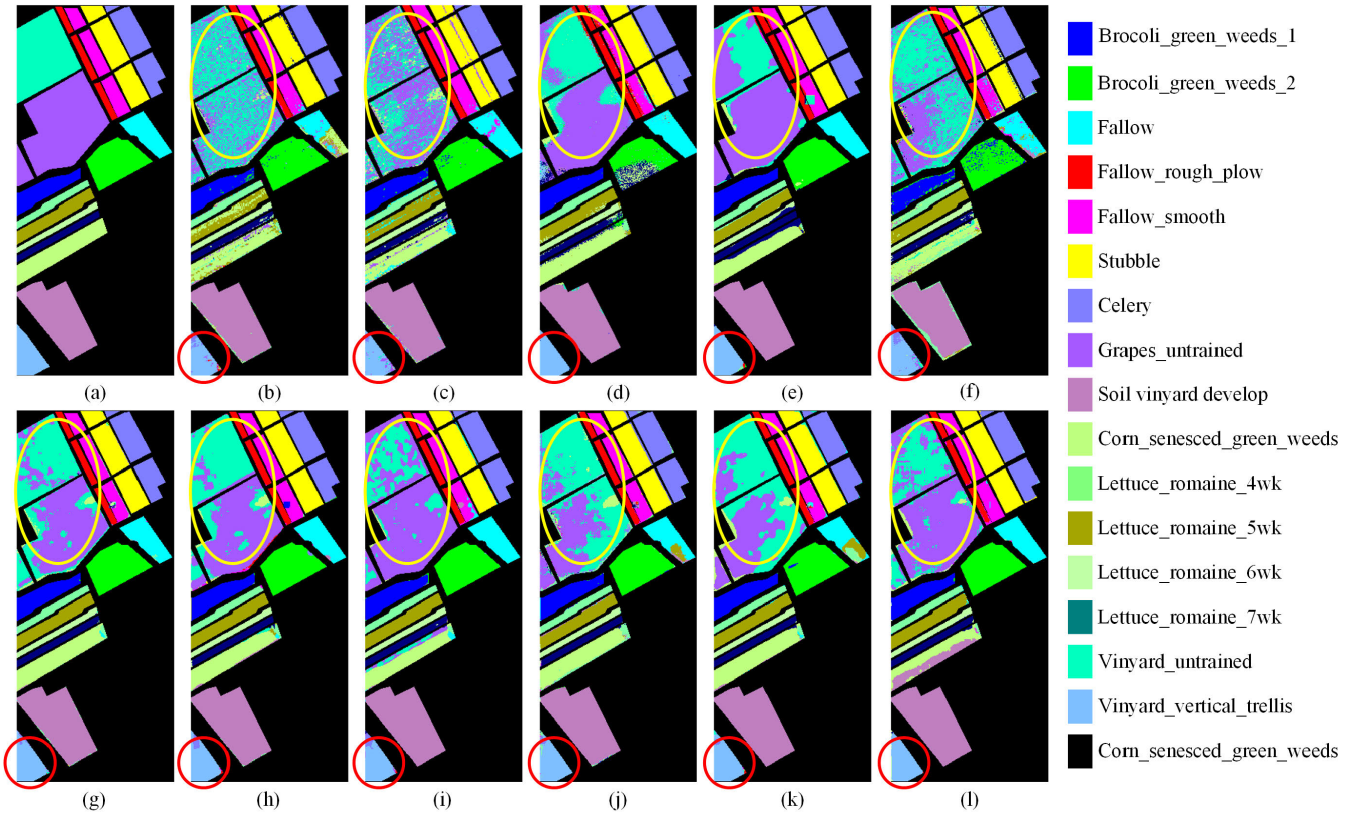


Fig. 14. Classification map on the SA dataset. (a) Ground truth. Classification maps via various algorithms for the SA dataset. (b) KNN, (c) SVM, (d) HybridSN, (e) SSFTT, (f) DFSL, (g) DCFSL, (h) Gia-CFSL, (i) CMFSL, (j) ACDFSL, (k) RPCL-FSL, and (l) LMFSL.

negative impacts caused by domain shifts. To capture nonlocal information more effectively, Gia-CFSL combines graph information aggregation with a domain discriminant module. The CMFSL method improves the classification boundaries of the nonlinear discriminator by converting the Euclidean distance metric space into a category-covariance metric space, thereby enhancing the model's classification performance. The ACDFSL method, adds the SE attention mechanism to the DCFSL method to ensure the stability of the model channel dimension. RPCL-FSL enhances the stability of the prototype by adding triple constraints based on contrast learning, self-calibration, and cross-calibration to FSL.

To fairly compare the methods, all FSL algorithms use the same XA dataset as the source domain to ensure the consistency. The following conclusions can be obtained by analyzing Tables VI–IX.

3) Comparison of Non-FSL Methods: Among the four algorithms, KNN, SVM, HybridSN, and SSFTT, the first two methods of KNN and SVM demonstrate similar classification accuracies across different datasets. For deep learning methods, HybridSN exhibits superior classification accuracy over KNN and SVM on the UP, IP, and SA datasets. However, on the PC dataset, HybridSN slightly underperforms KNN and SVM, which may be due to its larger patch size setting that ignores the local information.

4) Comparison of FSL Methods: DCFSL, Gia-CFSL, CMFSL, ACDFSL, and RPCL-FSL all introduce their own improvements on the framework of DFSL, which results in a

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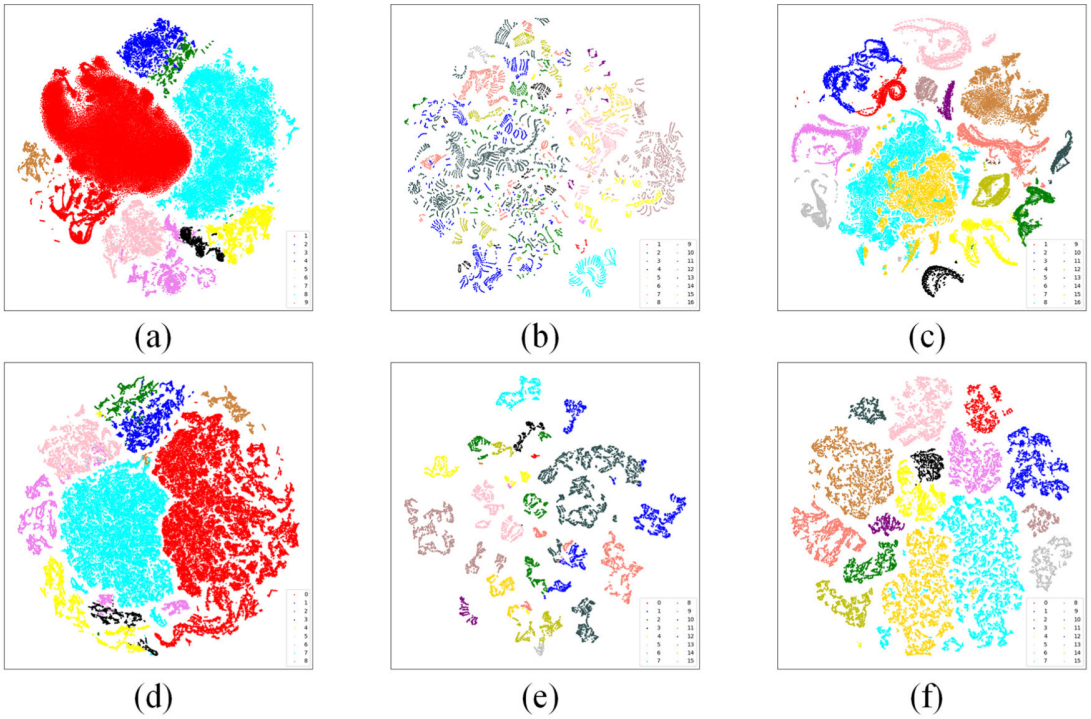


Fig. 15. Category separability of HSI features. (a) Original PC features. (b) Features learned by LMFSL on PC. (c) Original IP features. (d) Features learned by LMFSL on IP. (e) Original SA features. (f) Features learned by LMFSL on SA.

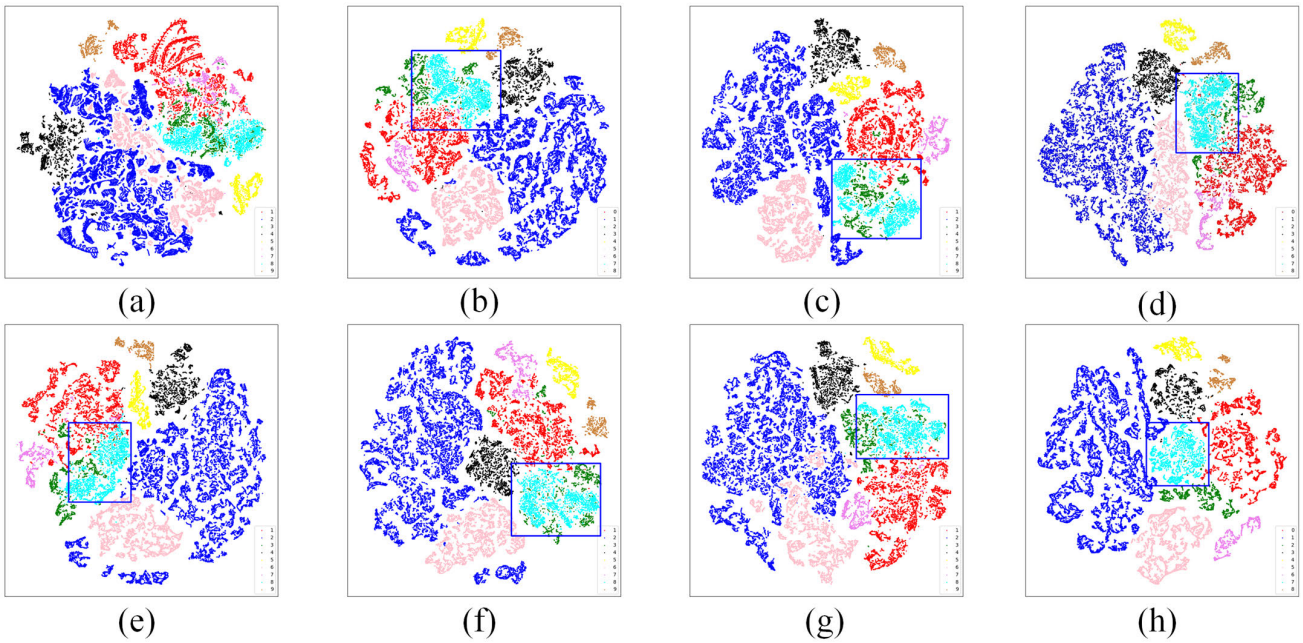


Fig. 16. Category separability of HSI features. (a) Original UP features. (b) Features learned by DFSL on UP. (c) Features learned by DCFSL on UP. (d) Features learned by Gia-CFSL on UP. (e) Features learned by CMFSL on UP. (f) Features learned by ACDFSL on UP. (g) Features learned by RPCL-FSL on UP. (h) Features learned by LMFSL on UP.

certain improvement in classification performance over DFSL. Specifically, DCFSL incorporates a domain discrimination module based on adversarial training to mitigate the impacts of domain shift. Gia-CFSL further enhances the ability of capturing the nonlocal information by incorporating a graph information aggregation technique in the domain discrimination module. CMFSL transforms the original Euclidean distance metric space into a category-covariance metric space to reduce the misclassification. ACDFSL method introduces the SE attention mechanism to increase the stability of the model channel dimension. RPCL-FSL combines FSL and contrastive learning and imposes triple constraints on it, thus enhancing the stability of the support set prototype. All these methods adopt the strategy of alternating the training metric space in order to minimize the negative impacts of domain shifts, thus effectively enhancing the classification accuracy of the target domain.

5) FSL Methods Versus Non-FSL Methods: Considering all evaluation metrics, FSL methods generally outperform non-FSL algorithms, though there are exceptions under

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TABLE VIII
CLASSIFICATION RESULT (%) ON THE IP DATASET (FIVE LABELED SAMPLES EACH CATEGORY)

Class	KNN	SVM	HybridSN	SSFTT	DFSL	DCFSL	Gia-CFSL	CMFSL	ACDFSL	RPCL-FSL	LMFSL
1	69.02±10.35	54.88±25.24	90.73±6.43	99.76±0.73	90.49±7.67	94.15±9.65	88.78±16.87	98.54±1.62	92.44±10.54	**99.27±1.12**	98.05±2.39
2	28.15±6.42	24.61±11.17	22.26±5.68	48.29±5.94	38.48±7.22	47.03±13.15	36.36±10.54	48.80±6.83	45.66±9.82	52.21±9.77	**65.66±10.39**
3	32.92±8.81	37.66±8.32	46.68±10.65	35.48±9.88	34.98±8.99	54.52±3.56	43.08±8.10	59.78±8.53	50.01±12.60	54.27±10.17	**64.24±12.51**
4	35.43±9.80	42.11±9.27	56.77±8.43	86.29±12.04	56.42±16.46	82.84±11.62	75.17±9.30	83.41±13.32	74.91±18.46	81.34±17.27	**89.70±5.40**
5	65.10±9.11	55.23±17.79	**80.13±4.35**	82.82±3.57	67.93±6.45	74.16±6.23	69.67±7.55	76.97±7.48	72.28±6.52	76.21±7.56	79.23±7.94
6	68.95±6.64	59.94±8.65	79.72±5.62	**93.85±3.05**	78.66±6.20	86.54±5.94	78.77±11.90	78.80±11.04	88.55±5.11	89.50±5.83	93.08±3.52
7	86.96±9.33	78.70±13.25	96.52±5.43	**100.00±0.00**	95.22±7.89	98.26±3.98	95.65±8.25	99.13±1.74	99.13±2.61	98.70±3.91	**100.00±0.00**
8	49.56±12.40	45.26±14.59	89.81±8.30	72.39±12.92	81.46±10.28	86.43±13.04	88.77±6.54	87.17±15.29	88.96±14.72	**90.13±11.05**	87.29±16.23
9	80.67±10.93	88.67±11.94	**100.00±0.00**	98.67±2.67	98.00±3.06	**100.00±0.00**	98.67±4.00	99.33±2.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	41.81±11.15	37.94±6.74	60.75±5.83	**82.33±4.81**	51.31±9.21	60.81±8.23	62.35±12.16	66.45±6.78	61.48±8.03	67.96±9.15	76.86±7.07
11	34.73±8.67	36.04±10.44	30.50±6.03	49.68±4.71	50.22±11.86	58.99±11.58	62.27±10.01	58.08±15.37	59.65±14.64	62.54±9.07	**69.75±6.22**
12	28.89±6.30	29.98±10.43	42.09±5.96	50.51±11.68	33.69±6.55	45.14±12.03	40.88±9.08	48.15±15.20	38.38±7.42	52.47±12.04	**56.65±14.32**
13	91.10±3.73	86.25±4.09	88.00±5.97	**100.00±0.00**	96.70±4.30	97.45±4.38	96.65±5.01	98.45±3.23	97.10±3.38	98.25±2.28	99.30±1.40
14	60.52±15.55	64.73±14.96	67.21±10.76	**93.29±4.28**	77.78±7.25	87.67±8.73	85.29±5.73	84.37±8.28	89.25±5.86	84.06±10.17	91.65±6.44
15	23.15±4.24	36.51±11.96	50.03±17.20	78.79±9.24	59.84±12.41	76.75±10.52	68.35±12.11	72.44±13.45	70.89±13.06	83.31±7.22	**87.82±9.15**
16	87.61±2.61	46.48±10.41	99.77±0.45	99.77±0.68	95.80±3.97	98.41±2.23	97.50±3.48	99.55±0.7	**99.89±0.35**	99.66±0.73	98.52±2.28
OA	43.22±2.95	42.48±2.78	50.91±2.71	66.39±1.78	56.36±3.13	66.43±2.89	63.21±2.65	66.70±2.38	65.68±3.91	69.39±2.48	**76.48±2.18**
AA	55.29±1.96	51.56±1.87	68.81±2.40	79.50±1.32	69.19±2.62	78.07±1.78	74.26±2.84	78.71±0.97	76.79±2.72	80.62±1.62	**84.86±1.69**
Kappa	36.84±3.12	36.10±2.65	46.43±2.89	62.48±1.93	51.08±3.26	62.30±3.10	58.56±2.95	62.54±2.33	61.43±4.18	65.52±2.52	**73.46±2.43**

TABLE IX
CLASSIFICATION RESULT (%) ON THE SA DATASET (FIVE LABELED SAMPLES EACH CATEGORY)

Class	KNN	SVM	HybridSN	SSFTT	DFSL	DCFSL	Gia-CFSL	CMFSL	ACDFSL	RPCL-FSL	LMFSL
1	95.57±2.29	95.35±2.44	99.32±0.68	**99.97±0.10**	95.38±3.86	99.83±0.28	98.32±1.58	97.75±2.42	99.80±0.41	99.74±0.43	99.45±0.67
2	97.46±2.67	93.33±4.05	94.69±7.45	**99.98±0.05**	95.85±2.21	99.37±1.09	99.42±0.56	99.39±1.03	99.85±0.17	99.74±0.53	99.97±0.04
3	77.60±10.58	87.67±7.83	95.27±5.89	**99.79±0.27**	92.09±4.20	92.60±8.43	92.18±9.80	94.58±9.28	91.53±9.33	89.47±9.00	96.32±6.69
4	97.85±2.05	97.48±2.76	99.22±0.50	97.33±0.56	99.37±0.77	99.52±0.55	99.32±0.46	99.31±0.59	98.84±1.47	**99.63±0.30**	97.53±3.04
5	92.60±4.71	92.85±5.53	89.39±7.35	93.43±1.45	87.10±5.96	91.47±5.65	92.71±3.73	93.28±2.64	91.46±4.22	92.79±2.59	**95.77±1.98**
6	97.90±1.32	89.95±8.08	97.73±2.46	**99.79±0.28**	98.39±1.36	99.56±0.42	99.10±1.57	99.24±1.15	99.49±0.89	99.03±0.69	99.30±1.59
7	99.34±0.21	98.39±0.93	96.96±3.63	97.75±1.40	96.01±4.12	98.79±0.98	98.18±2.23	98.17±2.61	97.99±2.43	98.87±1.63	**99.48±0.51**
8	45.31±7.73	51.74±9.28	75.56±5.12	56.66±10.21	70.33±7.80	76.65±8.43	74.97±5.91	70.24±15.35	74.84±8.98	75.85±8.83	**82.61±5.98**
9	97.27±1.03	96.09±1.52	95.34±4.36	99.63±0.43	93.25±3.12	**99.75±0.30**	98.50±1.58	99.45±0.95	99.15±0.96	99.72±0.36	99.38±1.21
10	72.03±7.52	72.76±8.95	**95.09±3.57**	91.95±3.43	72.76±8.59	87.28±7.38	84.85±4.28	86.20±8.17	87.28±5.80	84.48±13.42	92.40±4.75
11	89.17±2.34	87.16±4.25	94.20±3.89	99.75±0.24	89.79±6.91	98.94±1.19	96.49±3.09	98.09±1.55	**99.82±0.78**	98.11±2.34	99.47±0.64
12	89.97±7.44	93.23±4.28	94.99±3.14	94.12±4.92	96.26±3.91	**99.64±0.33**	99.26±0.70	98.57±3.53	98.93±0.76	99.44±1.50	98.92±1.22
13	96.60±2.41	96.88±2.36	97.10±2.40	96.41±5.14	98.47±1.38	99.40±0.58	99.42±0.53	99.31±1.39	99.21±0.75	99.46±0.33	**99.76±0.33**
14	90.73±2.79	87.30±5.96	98.23±1.13	88.22±5.42	97.39±1.67	**99.20±0.73**	98.07±1.49	98.97±0.63	99.17±0.96	98.64±1.65	96.63±5.76
15	60.13±10.93	61.13±11.83	73.33±12.55	**94.30±2.29**	72.27±5.15	76.22±10.27	79.46±6.73	74.83±10.39	78.27±5.48	78.86±5.80	83.23±11.03
16	85.32±7.63	88.11±9.09	**97.50±2.19**	96.66±2.45	86.74±5.65	93.12±4.36	87.41±6.30	89.51±8.43	92.72±6.85	90.78±4.56	96.81±3.28
OA	78.16±1.31	79.09±0.60	88.56±1.74	88.48±1.78	85.03±1.92	89.98±1.48	89.42±1.88	88.20±2.27	89.69±2.09	89.83±1.61	**92.85±1.06**
AA	86.55±1.43	86.84±1.13	93.37±1.23	94.11±0.49	90.09±1.15	94.46±1.13	93.60±1.39	93.56±1.07	94.22±1.18	94.04±0.86	**96.06±0.64**
Kappa	75.86±1.44	76.82±0.65	87.33±1.92	87.28±1.95	83.39±2.11	88.87±1.64	88.26±2.08	86.90±2.48	88.55±2.31	88.71±1.77	**92.05±1.18**

TABLE X
ARCHITECTURES WITH DIFFERENT MODULES

	LFE-MAs	CLSM	MSOM
LMFSL_L0	×	×	×
LMFSL_L1	√	×	×
LMFSL_L2	√	√	×
LMFSL_L3	√	×	√
LMFSL	√	√	√

specific conditions. For instance, the SSFTT method has surpassed the basic DFSL framework in classification performance by integrating hybrid convolution with transformer techniques. However, compared to algorithms like DCFSL, Gia-CFSL, CMFSL, ACDFSL, and RPCL-FSL, SSFTT behaves slightly insufficient classification performance. This is largely a consequence of the effects of domain shifts: when there is a significant difference in data distribution between the source and target domains, FSL algorithms are constrained in extracting transferable information. Accordingly, the algorithms of DCFSL, Gia-CFSL, CMFSL, ACDFSL, and RPCL-FSL alleviate this problem to a certain extent, thus achieving higher classification accuracy.

6) Performance Advantages of LMFSL: Compared with other FSL methods, the LMFSL proposed in this article exhibits extremely high classification accuracy on UP, PC, IP, and SA datasets, with the OAs that are higher than the best results of the compared models by 4.89%, 0.02%, 7.09%, and 2.87%, respectively. This is mainly due to the joint use of the proposed CLSM, MSOM, and LFE-MAs, which makes the consistency metric space mapped by LMFSL more category—differentiable and thus significantly improves the recognition performance of the model. To intuitively compare the classification effects of different methods, Figs. 11–14 show the ground-truth maps and classification maps of the various methods on the four test datasets, respectively, where

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the classification maps represent the best results among the ten experiments. By carefully comparing and analyzing the classification maps, it can be found that the FSL method has obvious advantages over the non-FSL method. And it is closer to the ground truth conditions, For example, for category 8 in the UP dataset, category 5 in the PC dataset, category 11 in the IP dataset, and category 8 in the SA dataset, respectively. In particular, the classification maps generated by LMFSL are the closest to the ground-truth maps.

To further highlight the category differentiation performance of LMFSL, Fig. 15 illustrates the distribution of the original features on the three datasets, PC, IP, and SA, with the features processed by LMFSL in a 2-D space. The results show that LMFSL performs well in terms of category discriminability. Fig. 16 further shows the visualization of the original features of the UP dataset and the embedding features extracted by each FSL method downscaled to the 2-D plane, from which it is easy to see that LMFSL has the best ability to discriminate between categories.

D. Ablation Study

To systematically evaluate the performance contribution of each component module in the proposed LMFSL network, this study incrementally adds different modules to the baseline network, defining different networks of varying complexity, as shown in Table X. The necessity of each module will be analyzed by examining the categorized performance before and after the addition of each module.

1) LMFSL_L0: A baseline network that does not contain the MSOM and the CLSM shown in Fig. 2, but it uses the same feature extraction module as DFSL, DCFSL, Gia-CFSL, ACDFSL, and RPCL-FSL, while it employs the Euclidean distance as a similarity metric.

2) LMFSL_L1: It represents an enhancement over the baseline network by replacing the feature extractor with LFE-MAs, which contains MAFE blocks that can effectively mine both local and nonlocal information. This makes it easier to extract more distinguishable embedded features, thus improving the model's classification performance.

3) LMFSL_L2: It represents an advancement of LMFSL_L1 with the addition of CLSM. As the previous model was affected by noise to a greater extent, it caused instability in the model's classification performance. The addition of CLSM can introduce a certain degree of uncertainty, which helps prevent the model from being overconfident, thus effectively improving the model's noise resistance.

4) LMFSL_L3: It indicates that MSOM is further added to LMFSL_L1. In view of the confusion in the distribution of samples in the metric space, this module optimizes the distribution of samples in the metric space by increasing the aggregation degree of the same category and refining the decision boundaries of category discrimination, which will effectively reduce the misclassification among the categories.

5) LMFSL: It is the LMFSL framework proposed in this study, which is based on LMFSL_L1, and incorporates both CLSM and MSOM. As it combines the advantages of LMFSL_L2 and LMFSL_L3, thus further improving the performance of the network model.

TABLE XI

CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT MODULES ON THE UP DATASET (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Class	LMFSL-L0	LMFSL-L1	LMFSL-L2	LMFSL-L3	LMFSL
1	77.41±8.44	**89.73±5.59**	84.36±7.38	89.30±4.42	88.52±6.46
2	86.86±6.95	89.11±5.89	90.48±7.04	89.20±6.21	**91.29±5.28**
3	64.26±15.02	70.94±10.11	**73.60±8.51**	66.44±11.55	72.20±10.53
4	**94.17±2.46**	90.55±3.40	90.51±3.32	89.18±4.72	92.39±2.76
5	**99.81±0.24**	99.33±0.58	99.18±0.66	99.38±0.52	99.23±0.62
6	76.64±14.03	76.45±13.82	78.79±13.36	**81.29±13.24**	79.33±17.21
7	80.29±6.38	88.60±7.06	93.94±4.47	**94.60±3.78**	91.99±6.35
8	64.69±10.57	85.93±6.09	**86.02±4.60**	84.48±6.06	84.16±9.69
9	97.06±3.44	94.88±6.25	97.49±3.09	**98.35±1.33**	97.10±3.49
OA	81.13±3.22	86.09±3.17	87.48±2.85	87.45±2.04	**88.38±2.50**
AA	81.35±2.22	86.28±2.49	88.26±2.03	88.02±2.11	**88.47±2.96**
Kappa	76.81±3.91	82.59±4.03	83.61±3.53	83.58±2.55	**84.73±3.35**

TABLE XII

CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT MODULES ON THE PC DATASET (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Class	LMFSL-L0	LMFSL-L1	LMFSL-L2	LMFSL-L3	LMFSL
1	99.38±0.45	99.34±0.40	**99.76±0.30**	99.62±0.27	99.50±0.28
2	89.44±3.93	90.03±4.96	**93.38±6.38**	90.43±4.28	92.34±3.21
3	**90.01±6.29**	87.40±10.52	83.88±12.19	88.26±5.45	89.47±5.99
4	88.90±10.85	91.89±8.09	**94.97±4.67**	90.36±8.97	92.54±6.69
5	83.45±6.36	86.85±5.39	88.32±5.69	85.48±6.00	**91.55±5.00**
6	97.78±1.94	97.48±1.54	93.58±6.76	97.95±1.55	**98.55±1.50**
7	85.39±6.76	85.53±6.11	83.95±4.01	86.89±6.04	**86.92±5.14**
8	97.88±2.27	98.13±0.90	98.29±1.28	98.84±1.24	**99.04±0.74**
9	95.58±3.05	95.93±3.33	**97.89±0.99**	96.03±3.42	95.99±2.74
OA	96.48±0.86	96.71±0.52	96.89±0.85	97.09±0.62	**97.57±0.48**
AA	91.98±1.77	92.51±1.54	92.67±1.63	92.65±1.29	**93.99±1.26**
Kappa	95.04±1.20	95.36±0.73	95.60±1.20	95.89±0.86	**96.56±0.67**

Tables XI–XIV show the classification results of different ablation experiments for each dataset, and the following conclusions can be summarized after analysis.

1) By comparing LMFSL_L1 and LMFSL_L0, it is observed that after applying LFE-MAs, the OA of the four datasets is increased by 4.96%, 0.23%, 5.07%, and 2.2%, respectively. This can be attributed to the fact that LFE-MAs simultaneously take into account the global and local features while integrating the shallow and deep features, resulting in a stronger feature extraction capability and thereby enhancing the classification performance of the model.
2) Comparing LMFSL_L2 and LMFSL_L1, the network models with the addition of CLSM all have improved the classification indicator of OA on all four datasets; and at the same time, the standard deviation of classification is reduced on all three datasets except the SA dataset. This is because CLSM reduces the probability of the correct category to assign it to the rest of the categories, which introduces a certain amount of uncertainty. It is the fact that the model incorporates such uncertainty into the learning that allows the model to become more noise-resistant, thus improving the robustness and stability of the model. In addition, the relationships between categories provided by CLSM act as a form of data

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TABLE XIII
CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT MODULES ON THE IP DATASET (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Class	LMFSL-L0	LMFSL-L1	LMFSL-L2	LMFSL-L3	LMFSL
1	94.15±6.74	97.32±5.82	97.07±2.84	**99.29±1.52**	98.05±2.39
2	48.41±10.19	56.42±9.79	63.84±11.64	64.96±9.60	**65.66±10.39**
3	53.25±11.96	**68.78±9.84**	65.33±12.55	65.15±8.55	64.24±12.51
4	82.84±9.25	84.14±13.26	87.72±6.36	**91.76±8.85**	89.70±5.40
5	72.53±5.47	**82.93±2.74**	76.65±9.08	77.08±6.99	79.23±7.94
6	87.05±4.84	90.46±7.62	88.90±7.64	90.63±6.01	**93.08±3.52**
7	99.57±1.30	99.57±1.30	99.57±1.30	99.58±1.25	**100.00±0.00**
8	86.66±10.09	**88.46±13.47**	84.57±12.90	88.25±14.84	87.29±16.23
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	61.19±5.47	69.60±6.20	74.07±7.46	74.96±8.51	**76.86±7.07**
11	61.87±9.63	57.75±8.92	65.18±10.22	68.04±6.61	**69.75±6.22**
12	39.85±14.72	60.00±12.35	55.60±14.52	**62.02±15.53**	56.65±14.32
13	91.90±12.79	97.60±3.95	98.35±1.83	**99.50±0.70**	99.30±1.40
14	87.00±4.11	89.36±9.75	88.76±4.51	84.69±8.89	**91.65±6.44**
15	75.56±11.78	87.59±10.00	85.17±9.95	81.18±11.33	**87.82±9.15**
16	97.73±3.25	**99.09±1.75**	96.70±3.50	96.97±4.52	98.52±2.28
OA	66.68±3.37	71.75±3.59	73.80±2.07	74.88±2.33	**76.48±2.18**
AA	77.47±2.68	83.06±2.30	82.97±1.34	84.00±1.80	**84.86±1.69**
Kappa	62.49±3.70	68.40±3.93	70.51±2.22	71.75±2.57	**73.46±2.43**

TABLE XIV
CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT MODULES ON THE SA DATASET (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Class	LMFSL-L0	LMFSL-L1	LMFSL-L2	LMFSL-L3	LMFSL
1	98.22±1.99	99.05±0.85	**99.72±0.27**	98.99±1.14	99.45±0.67
2	99.85±0.18	99.88±0.18	99.85±0.28	99.14±1.59	**99.97±0.04**
3	93.92±10.21	95.95±3.89	90.33±13.51	86.53±14.18	**96.32±6.69**
4	**98.72±2.53**	97.52±2.23	98.26±1.01	98.24±1.74	97.53±3.04
5	91.70±3.80	95.11±2.85	95.52±2.22	92.67±5.18	**95.77±1.98**
6	99.83±0.29	99.65±0.88	**99.84±0.38**	99.05±1.06	99.30±1.59
7	99.40±0.70	98.33±1.67	99.18±0.88	99.19±1.21	**99.48±0.51**
8	75.78±7.75	81.53±4.94	81.02±5.36	80.75±4.38	**82.61±5.98**
9	99.20±0.99	99.25±0.99	**99.50±0.46**	98.79±1.70	99.38±1.21
10	88.26±5.99	88.30±10.61	90.98±3.77	91.93±3.72	**92.40±4.75**
11	97.46±2.80	98.67±1.18	99.28±1.00	**99.48±0.48**	99.47±0.64
12	99.53±0.83	**99.60±0.39**	99.07±0.80	99.13±1.65	98.92±1.22
13	99.46±0.47	99.29±0.69	99.59±0.44	99.22±1.07	**99.76±0.33**
14	99.11±0.62	97.77±3.18	99.24±0.86	**99.35±0.72**	96.63±5.76
15	74.32±5.90	80.01±10.10	82.06±9.37	**84.62±6.92**	83.23±11.03
16	93.44±7.43	95.60±5.65	95.21±6.01	96.22±4.86	**96.81±3.28**
OA	89.59±1.05	91.79±1.19	92.09±1.19	91.98±1.26	**92.85±1.06**
AA	94.26±0.63	95.34±1.10	95.54±1.12	95.21±0.92	**96.06±0.64**
Kappa	88.44±1.16	90.87±1.34	91.22±1.32	91.10±1.39	**92.05±1.18**

augmentation for FSL tasks, as well as reducing feature normalization and facilitating sample clustering, which further improves the generalization of the model.

3) Comparing LMFSL_L3 with LMFSL_L1, the OA of each dataset is substantially improved by adding MSOM. MSOM optimizes the sample distribution by maximizing the intraclass similarity and minimizing the interclass similarity, which further strengthens the degree to which the same kind of data focuses on their inherent prototypes, and effectively improves the classification decision boundaries of the categories within the metric space. Consequently, it enables learning of the embedding features of the target domain with fewer labeled samples, mitigates misclassification scenarios, and endows the model with stronger discriminative capabilities.

4) By combining the advantages of LMFSL_L2 and LMFSL_L3, LMFSL achieves the best classification performance in terms of OAs upon all four datasets. This is mainly due to the fact that LMFSL applies LFE-MAs as the feature extractor, which effectively improves the feature extraction capability. Meanwhile, the module of CLSM makes LMFSL more noise-resistant and improves the generalization ability of the model. Moreover, the further addition of the MSOM refines the decision boundaries of the categories so that the model reduces category misclassification.

E. Further Analysis of Parameter Settings

To investigate the impact of hyper-parameters and network structure in the experimental setup on the classification results of the model, some necessary related experiments and discussions will be carried out in this section.

1) Exploration of HSI Cube Size: The input data used in this study consists of hyperspectral cubes with well-defined spatial extents centered on pixels of specific categories. Due to the significant differences in the background environments in which the samples are embedded, the size of the spatial neighborhood thus becomes a key variable affecting the classification performance and computational efficiency. To specify the optimal input spatial neighborhood size, this study designs a series of experiments to quantify the impact of different spatial neighborhood sizes $s = \{5, 7, 9, 11, 13\}$, on the classification accuracy and model training time. The experimental results, as shown in Table XV, indicate that in all test datasets except the SA dataset, the model achieves the highest classification accuracy while maintaining relatively low computational costs when the spatial neighborhood size was set to 9. Combining the above experimental results, LMFSL is able to achieve the best balance of performance and computational efficiency for the model when using a spatial neighborhood of size 9 as input.

2) Selection of the Optimal Number of MAFE Blocks in LFE-MAs: In the proposed feature extractor LFE-MAs, the number of MAFE blocks is identified as a factor that has a critical impact on the classification performance. To scientifically quantify its impact on the model performance, a set of systematic experiments are designed in this study while controlling the other hyper-parameters to be constant. The experimental results, as detailed in Table XVI, reveal a clear trend: as the number of MAFE blocks increases, the OA of the model roughly shows a pattern of increasing and then decreasing. Particularly, when the number of MAFE blocks is set to 3, the model achieves the optimal classification performance on three test datasets. Based on the experimental results, three MAFE blocks are finally selected within the LFE-MAs framework to achieve the best classification performance.

3) Influence of Learning Rates: The learning rate plays a key role in the feature learning and convergence speed of a model. A learning rate that is too small can lead to a significant increase in model training time, while a learning rate that is too large may prevent the model from converging properly, thus negatively affecting the model's performance. To accurately assess the role of learning rate in LMFSL modeling, this study designs a set of experiments for three specific learning

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TABLE XV
CLASSIFICATION RESULTS (%) AND TRAINING TIME (S) OF DIFFERENT SPATIAL NEIGHBORHOOD SIZES

Input Size		s =5	s =7	s =9	s =11	s =13
UP	OA	83.02±3.69	85.86±3.30	**88.38±2.50**	87.72±2.55	87.79±2.02
	AA	85.58±2.55	86.27±2.19	**88.47±2.96**	87.76±2.68	86.96±2.61
	Kappa	77.92±4.49	81.44±4.20	**84.73±3.35**	83.88±3.31	84.02±2.57
	Training Time	290.57	315.50	375.81	439.40	719.68
PC	OA	96.94±0.54	97.18±0.46	**97.57±0.48**	96.82±0.39	96.48±0.83
	AA	93.62±0.67	**94.01±0.97**	93.99±1.26	92.99±1.12	92.25±1.29
	Kappa	95.68±0.76	96.02±0.65	**96.56±0.67**	95.51±0.55	95.04±1.16
	Training Time	381.54	445.82	485.00	559.35	693.53
IP	OA	67.12±4.02	70.19±3.67	**76.48±2.18**	72.86±2.97	72.61±3.45
	AA	79.05±2.02	81.28±2.68	**84.86±1.69**	82.62±2.09	83.00±1.83
	Kappa	63.1475±4.31	66.57±3.96	**73.46±2.43**	69.59±3.24	69.35±3.67
	Training Time	529.40	612.22	906.00	1074.66	1122.91
SA	OA	89.49±2.01	91.07±1.87	92.85±1.06	93.19±1.17	**93.21±1.45**
	AA	94.31±0.84	95.03±0.97	**96.06±0.64**	95.99±0.97	95.66±1.26
	Kappa	88.33±2.20	90.08±2.06	92.05±1.18	92.43±1.30	**92.45±1.61**
	Training Time	445.76	595.76	806.27	1139.42	1192.57

TABLE XVI
(%) OF DIFFERENT NUMBER OF MAFE BLOCKS ON THE FOUR DATASETS

Num.	1	2	3	4	5
UP	**88.81 ±2.31**	88.17 ±2.34	88.38 ±2.50	86.21 ±4.47	86.97 ±3.60
PC	97.46 ±0.69	97.51 ±0.49	**97.57 ±0.48**	97.19 ±0.61	96.97 ±0.74
IP	73.98 ±4.01	72.32 ±3.58	**76.48 ±2.18**	71.20 ±3.27	70.76 ±2.47
SA	92.33 ±1.48	92.57 ±0.93	**92.85 ±1.06**	91.89 ±0.89	92.01 ±0.78

rate settings (i.e., 0.01, 0.001, and 0.0001). The experimental results listed in Table XVII clearly show that a learning rate setting of 0.001 leads to the highest classification accuracy on all test datasets. Based on this finding, the learning rate parameter of the LMFSL model is set to 0.001 in this study to achieve the best performance of the model.

4) Determination of the Smoothing Parameters: The label smoothing factor plays a central role in CLSM, as evidenced by its ability to improve the model's resistance to noise by introducing a small amount of uncertainty. However, improper setting of the smoothing factor may have two negative effects: first, too high a smoothing factor may make the model too conservative, with less confidence in the correct category and resulting in the model being insufficient to learn specific, smoothing factor may cause the model to remain overconfident and thus less resistant to noise. To precisely determine the optimal value of the smoothing factor ε, a set of experiments on four datasets are designed in this study to quantify the impact of different smoothing factors on classification accuracy. The experimental results, as shown in Fig. 17, indicate that the model achieves the highest classification accuracy on each test dataset with the smoothing factor ε set to 0.1. Based on this, the smoothing factor ε is finally set to 0.1 in this study to enable the model to reach its optimal performance.

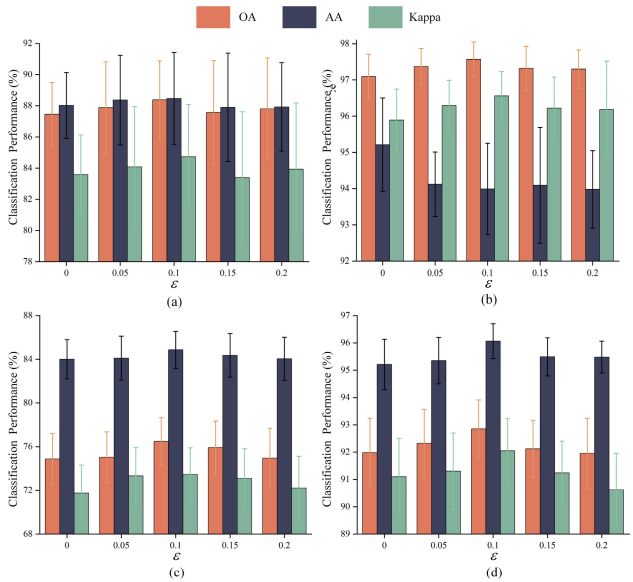


Fig. 17. Classification performance (%) corresponding to different smoothing parameters ε on four datasets. (a) UP, (b) PC, (c) IP, and (d) SA.

5) Discussion of Optimization Parameters: The proposed metric optimization module includes two parameters, the relaxation factor m and the maximum scale factor γ, which are used to tune the performance of the module. The parameter m dominates the expansion and contraction of the decision boundary, which in turn affects the distribution of distances between all samples within and outside each category. A larger value of m can lead to over-expansion of the decision boundary and increase the classification error. On the contrary, a smaller m may reduce the model's ability to discriminate between different categories. The parameter γ affects the model's focus on samples that are difficult to classify: too large a γ

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TABLE XVII
CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT LEARNING RATES ON THE FOUR DATASET (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Learning Rate	0.01			0.001			0.0001		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
UP	84.31±2.27	84.28±1.87	79.50±2.75	**88.38±2.50**	**88.47±2.96**	**84.73±3.35**	87.20±2.54	88.16±2.32	83.20±3.33
PC	96.91±0.84	93.59±1.35	95.64±1.18	**97.57±0.48**	**93.99±1.26**	**96.56±0.67**	97.05±0.75	93.53±1.90	95.84±1.06
IP	70.46±3.80	81.03±2.43	66.90±4.04	**76.48±2.18**	**84.86±1.69**	**73.46±2.43**	71.89±3.86	82.74±2.07	68.49±4.13
SA	91.14±1.23	94.93±1.12	90.16±1.37	**92.85±1.06**	**96.06±0.64**	**92.05±1.18**	91.70±1.17	95.43±0.76	90.78±1.28

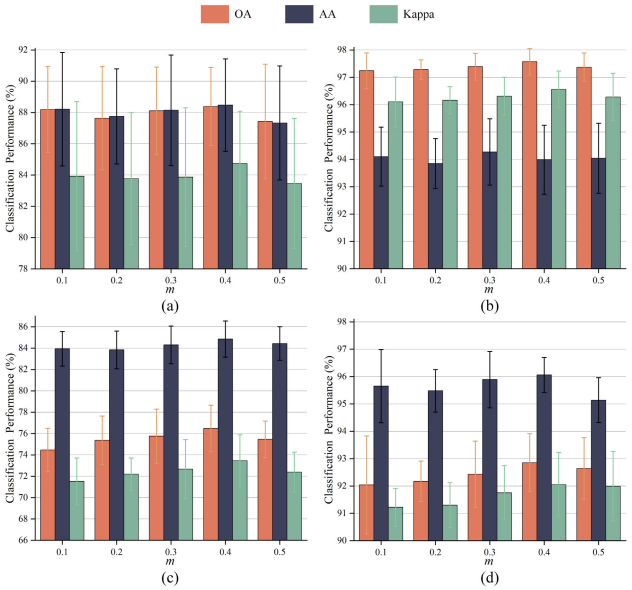


Fig. 18. Classification performance (%) corresponding to different relaxation factor m on four datasets. (a) UP, (b) PC, (c) IP, and (d) SA.

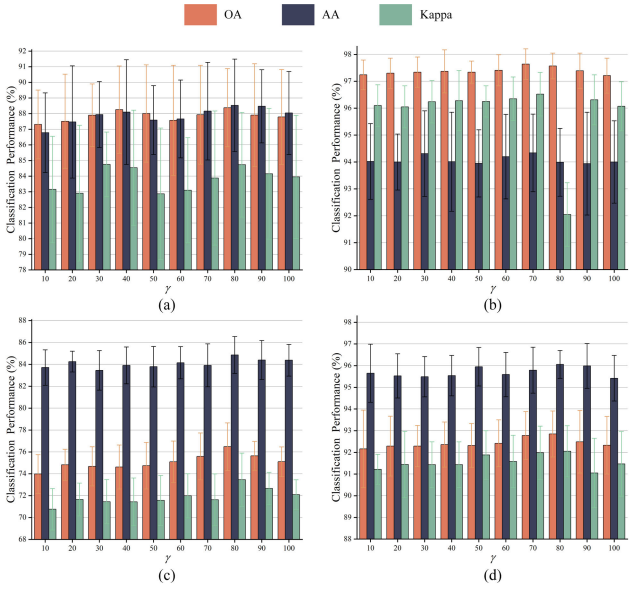


Fig. 19. Classification performance (%) corresponding to different maximum scale factor γ on four datasets. (a) UP, (b) PC, (c) IP, and (d) SA.

can trigger the model overfitting, while too small a γ can reduce the model's sensitivity to difficult-to-classify samples. To explore the optimal parameter settings, this study compares the classification performance of models under different parameter scenarios, as shown in Figs. 18 and 19. It can be observed that the model achieves the best performance on four datasets when m is set to 0.4; when γ is set to 80, the model achieves the best performance on three datasets and suboptimal performance on the rest dataset. For optimal classification performance, this method sets m to 0.4 and γ to 80.

6) Effects of the Number of Labeled Samples: All the above experiments are conducted under the scenario of selecting five labeled samples for each category in the test datasets. To further validate the effectiveness of the proposed method, this study carries out multiple rounds of experiments by varying the number of labeled samples (1, 2, 3, 4, and 5). The experiments are conducted upon four different datasets and OA is employed as the indicator for measuring the classification performance. As can be seen in Fig. 20, the experimental results show that the algorithms exhibit progressively enhanced classification accuracy as the number of labeled samples increase. Notably, the method proposed in this study achieves the best classification performance in most scenarios, and even in the few scenarios where the optimal performance is not achieved, it performs at the top of all methods. This result further confirms the superiority and applicability of the proposed method.

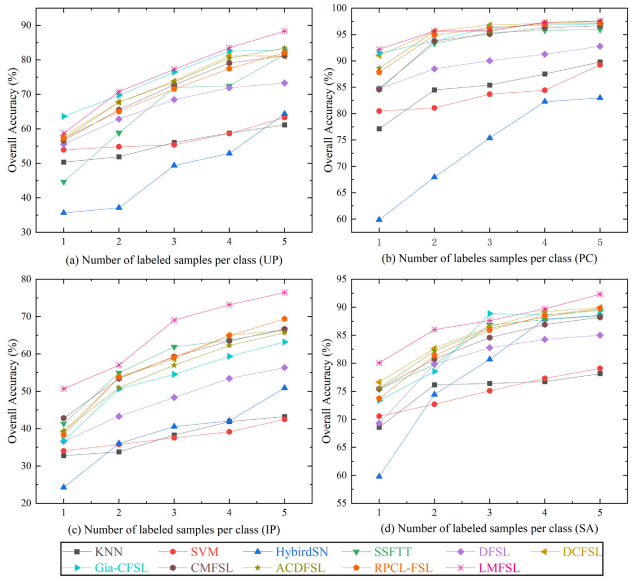


Fig. 20. OA with different labeled samples on four datasets. Number of labeled samples per class (a) UP, (b) PC, (c) IP, and (d) SA.

F. Analysis of the Computational Complexity

In large-scale HSI classification scenarios, the computational complexity is one of the key considerations in algorithm

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TABLE XVIII
MODEL COMPLEXITY (M, MILLION) AND EXECUTION TIME (SECONDS) OF DIFFERENT METHODS (FIVE LABELED SAMPLES FOR EACH CATEGORY)

Method			SVM	KNN	HybridSN	SSFTT	DFSL	DCFSL	Gia-CFSL	CMFSL	ACDFSL	RPCL-FSL	LMFSL
Model Complexity		Params	—	—	5.12	0.17	41.13	46.19	4518.82	0.62	79.29	46.59	3.43
		FLOPs	—	—	247.69	12.70	0.03	4.26	0.61	0.10	4.31	4.26	0.05
Execution Time	UP	Training	0.0006	0.0003	2.37	2.47	478.36	584.38	1403.15	492.01	1118.74	566.19	375.81
		Testing	0.0002	0.0094	2.77	1.19	3.61	3.374	3.09	6.02	4.52	4.61	3.78
	PC	Training	0.0008	0.0005	2.47	6.56	521.75	669.74	2036.94	553.32	1235.73	569.54	485.00
		Testing	0.0004	0.0123	10.02	5.04	13.25	12.22	14.09	3.17	18.37	17.21	15.06
	IP	Training	0.0015	0.0001	4.02	3.21	812.57	1120.71	2928.97	838.06	1539.62	860.10	806.27
		Testing	0.0005	0.0002	0.63	0.50	0.98	1.04	0.84	0.46	1.15	1.13	1.07
	SA	Training	0.0015	0.0113	2.03	3.78	874.23	1151.43	3094.20	732.27	1596.39	1036.41	906.00
		Testing	0.0006	0.0104	3.34	1.52	6.32	6.02	6.03	2.08	6.27	6.91	6.12

selection and design. To comprehensively evaluate the computational efficiency of the proposed LMFSL model, the study compares the computational costs for various classification methods, including training time, testing time, number of parameters, and floating-point operations (FLOPs). The experimental results shown in Table XVIII reveal the following two points.

1) FSL methods are generally more complex than non-FSL algorithms in terms of parameter size. However, the non-FSL algorithms perform better in terms of FLOPs, training, and testing time.
2) Compared to other FSL algorithms, the LMFSL model in this study has the smallest FLOPs and suboptimal parameter sizes, and achieves the lowest training time on all three datasets.

Taken together, the LMFSL model proposed in this study exhibits comprehensive advantages in terms of computational efficiency. Although LMFSL may be deficient in training and testing time compared to non-FSL algorithms, however, its significant advantage in classification accuracy cannot be ignored. And compared to other FSL methods, although LMFSL is slightly deficient in testing time, its significant advantages in model parameter sizes, FLOPs, and training time emphasize its relative superiority in terms of computational complexity. Therefore, there are good reasons to believe that LMFSL outperforms other FSL methods in terms of overall computational efficiency.

G. Influence of Different Source Domain

To further explore the effect of different source domains on LMFSL, this study conducted comparative experiments upon the Chikusei dataset and the Huston dataset. The Chikusei dataset contains 19 classes with a spatial extent of 2517 × 2335 pixels along with 128 bands; the Houston dataset contains 15 categories consisting of 349 × 1905 pixels and 144 bands. The experimental results are shown in Table XIX, and it can be seen that LMFSL outperforms the Huston dataset on the Chikusei dataset, which is mainly due to the fact that the Chikusei dataset contains 19 categories, which is larger than the 15 categories of the Huston dataset and ensures the extensiveness of the migration of the FSL process. The best results were achieved on the XA dataset, which contains 20 categories with a sufficient number of available samples in each category to avoid the imbalance of training samples.

TABLE XIX
CLASSIFICATION PERFORMANCE (%) OF LMFSL ON DIFFERENT SOURCE AND TARGET DATASETS

Source Dataset		UP	PC	IP	SA
XA	OA	**88.38±2.50**	**97.57±0.48**	**76.48±2.18**	**92.85±1.06**
	AA	**88.47±2.96**	**93.99±1.26**	84.86±1.69	**96.06±0.64**
	Kappa	**84.73±3.35**	**96.56±0.67**	**73.46±2.43**	**92.05±1.18**
Chikusei	OA	87.71±2.41	97.36±0.54	75.12±2.9	92.83±1.38
	AA	87.76±2.02	93.67±1.49	**84.98±1.84**	95.78±0.78
	Kappa	83.85±3.05	96.13±0.75	72.06±3.13	92.03±1.53
Huston	OA	86.43±3.31	97.20±0.56	74.62±3.38	92.17±1.28
	AA	86.51±2.15	93.11±1.66	84.47±1.49	95.60±0.84
	Kappa	82.21±4.23	96.04±0.79	71.56±3.67	91.30±1.41

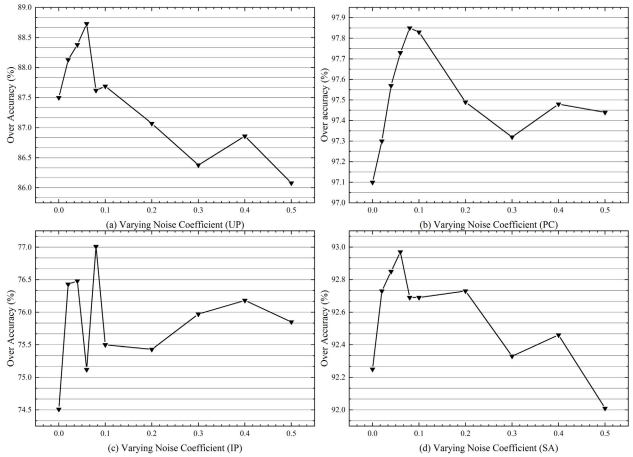


Fig. 21. Varying noise coefficients on four datasets (a) UP, (b) PC, (c) IP, and (d) SA.

Therefore, the more diverse and rich source domains are the more important for the FSL process of HSI classification.

H. Evaluation of Noise Immunity

The existing cross-domain FSL models usually take the approach of adding Gaussian noise with a certain proportion coefficient (the coefficient is 0.04) to labeled samples of the target domain dataset for data expansion, which poses a certain challenge to the model's noise immunity. The CLSM proposed by LMFSL combines with the FSL process to improve the model's noise immunity effectively.

To evaluate the noise immunity of LMFSL, this study carried out test experiments on each of the four datasets by adjusting the coefficients of the added Gaussian noise

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with other parameter settings unchanged, and the experimental results are shown in Fig. 21. It can be observed that adding a small amount of noise to the model realized data augmentation, which leads to a certain improvement of OA compared to no noise is added, but as the noise coefficient is further increased, the OA begins to fluctuate, but the trend of the change is relatively smooth and is able to maintain high accuracy. This finding indicates that the LMFSL has no tendency to learn the noise, effectively demonstrating the noise immunity of the proposed LMFSL.

V. Conclusion

In this study, a lightweight cross-domain few-shot HSI classification framework named LMFSL is constructed, which contains a CLSM for smoothing labels, an MOSM for optimizing the sample distribution, and a lightweight feature extractor LFE-MAs. The framework effectively solves some of the problems of current cross-domain FSL methods, such as weak noise immunity, difficulty in distinguishing the sample categories in the embedding metric space, and insufficient feature extraction for HSI, which gives LMFSL better generalization capabilities. Specifically, 1) the CLSM effectively enhances the category differentiation in the target domain and significantly improves the noise immunity of the model by combining the FSL tasks with label smoothing strategies. 2) The MSOM maximizes intraclass similarity and minimizes interclass similarity by assigning different weights to intraclass/interclass similarities, respectively. This promotes the aggregation of category prototypes and optimizes the distribution of samples in the metric space by maximizing intraclass similarity and minimizing interclass similarity by assigning different weights to intraclass/interclass similarities, respectively, to promote the aggregation of category prototypes, thereby effectively improving decision boundaries of the model. 3) LFE-MAs incorporate various attention mechanisms, enabling rapid and effective extraction of key features in HSI under limited sample conditions, thus enhancing the sample recognition capability of LMFSL. Through extensive experimental evaluations, the LMFSL framework significantly outperforms existing state-of-the-art methods on all four public HSI datasets, especially in terms of model complexity and computational efficiency. Since the domain shift among different data domains will have an impact on the robustness and generalization ability of the model, future research will mainly focus on how to mitigate the negative impact of domain shift to further improve the model performance in the target domain.

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